

Name: Anchal Singh

Roll No: 119

Module 1: Introduction to ASP.NET

1. Create a simple web page to show ASP.NET page Life cycle.

Default.aspx:

```
<%@Page Language="C#" AutoEventWireup="true" CodeBehind="Default.aspx.cs"
Inherits="PageLifeCycle.PageLifeCycle" %>
```

```
<!DOCTYPE html>
<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
<title>Page Life Cycle Demo</title>
</head>
<body>
<form id="form1" runat="server">
<asp:Label ID="lblStudentInfo" runat="server"
Text="Anchal Singh - Roll No: 119"
Font-Bold="True"
Font-Size="Large"></asp:Label>
<br /><br />
<div>
<asp:Button ID="btnSubmit" runat="server" Text="Submit"
OnClick="btnSubmit_Click" />
<br /><br />
<asp:Label ID="lblName" runat="server" Text=""></asp:Label>
</div>
</form>
</body>
</html>
```

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Default.aspx.cs:

```
using System;
using System.Web.UI;
namespace PageLifeCycle
{
    public partial class PageLifeCycle : System.Web.UI.Page
    {
        protected void Page_PreInit(object sender, EventArgs e)
        {
            lblName.Text = lblName.Text + "<br/>" + "PreInit";
        }

        protected void Page_Init(object sender, EventArgs e)
        {
            lblName.Text = lblName.Text + "<br/>" + "Init";
        }

        protected void Page_InitComplete(object sender, EventArgs e)
        {
            lblName.Text = lblName.Text + "<br/>" + "InitComplete";
        }

        protected override void OnPreLoad(EventArgs e)
        {
            lblName.Text = lblName.Text + "<br/>" + "PreLoad";
        }

        protected void Page_Load(object sender, EventArgs e)
        {
        }

        protected void btnSubmit_Click(object sender, EventArgs e)
        {
            lblName.Text = lblName.Text + "<br/>" + "btnSubmit_Click";
        }

        protected void Page_LoadComplete(object sender, EventArgs e)
        {
            lblName.Text = lblName.Text + "<br/>" + "LoadComplete";
        }

        protected override void OnPreRender(EventArgs e)
        {
            lblName.Text = lblName.Text + "<br/>" + "PreRender";
        }
    }
}
```

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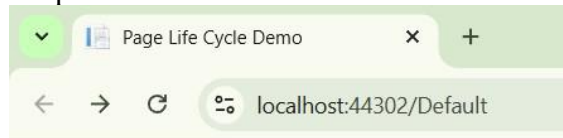
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```
}
```

```
protected override void OnSaveStateComplete(EventArgs e)
{
    lblName.Text = lblName.Text + "<br/>" + "SaveStateComplete";
}
```

```
protected void Page_UnLoad(object sender, EventArgs e)
{
    // This will not show on screen due to lifecycle
    lblName.Text = lblName.Text + "<br/>" + "UnLoad";
}
}
}
```

Output:



Anchal Singh - Roll No: 119

PreInit
Init
InitComplete
PreLoad
LoadComplete
PreRender
SaveStateComplete

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2. Design a web form to accept user input (e.g., name, email, and feedback) and display it after submission.

Default.aspx

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="RegistrationPage.aspx.cs"
Inherits="Experiment2.RegistrationPage" %>
```

```
<!DOCTYPE html>
<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
<title></title>
</head>
<body>
<form id="form1" runat="server">
<asp:Label ID="lblStudentInfo" runat="server"
Text="Anchal Singh - Roll No: 119"
Font-Bold="True"
Font-Size="Large"></asp:Label>
<br /><br />
<div>
</div>
<table class="auto-style1">
<tr>
<td>
<asp:Label ID="Label1" runat="server" Text="User Name"></asp:Label>
</td>
<td>
<asp:TextBox ID="username" runat="server" required="true"></asp:TextBox></td>
</tr>
<tr>
<td>
<asp:Label ID="Label6" runat="server" Text="Email ID"></asp:Label>
</td>
<td>
<asp:TextBox ID="EmailID" runat="server" TextMode="Email"></asp:TextBox></td>
</tr>
<tr>
<td>
<asp:Label ID="Label2" runat="server" Text="Password"></asp:Label></td>
<td>
<asp:TextBox ID="TextBox2" runat="server" TextMode="Password"></asp:TextBox></td>
</tr>
<tr>
<td>
<asp:Label ID="Label3" runat="server" Text="Confirm Password"></asp:Label></td>
```

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```
<td>
<asp:TextBox ID="TextBox3" runat="server" TextMode="Password"></asp:TextBox></td>
</tr>
<tr> <td>
<asp:Label ID="Label4" runat="server" Text="Gender"></asp:Label></td>
<td>
<asp:RadioButton ID="RadioButton1" runat="server" GroupName="gender" Text="Male" />
<asp:RadioButton ID="RadioButton2" runat="server" GroupName="gender" Text="Female"
/></td>
</tr>
<tr> <td>
<asp:Label ID="Label5" runat="server" Text="Select Course"></asp:Label>s</td>
<td>
<asp:CheckBox ID="CheckBox1" runat="server" Text="Advance Java" />
<asp:CheckBox ID="CheckBox2" runat="server" Text="DotNet" />
<asp:CheckBox ID="CheckBox3" runat="server" Text="Mobile Computingk" />
</td>
</tr>
<tr> <td>
</td>
<td> <br />
<asp:Button ID="Button1" runat="server" Text="Register" CssClass="btn btn-primary"
OnClick="Button1_Click"/>
</td>
</tr>
</table>
<asp:Label ID="message" runat="server" Font-Size="Medium"
ForeColor="Red"></asp:Label>
<table class="auto-style1">
<tr>
<td class="auto-style2"><asp:Label ID="UName" runat="server" ></asp:Label></td>
<td>
<asp:Label ID="ShowUserName" runat="server" ></asp:Label></td>
</tr>
<tr>
<td class="auto-style2"><asp:Label ID="Email" runat="server" ></asp:Label></td>
<td>
<asp:Label ID="UserEmailid" runat="server" ></asp:Label></td>
</tr>
<tr>
<td class="auto-style3"><asp:Label ID="Gender" runat="server" ></asp:Label></td>
<td class="auto-style4">
<asp:Label ID="ShowGender" runat="server" ></asp:Label></td>
</tr>
<tr>
<td class="auto-style2"><asp:Label ID="Course" runat="server" ></asp:Label></td>
```

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```
<td>
<asp:Label ID="ShowCourses" runat="server" ></asp:Label></td>
</tr>
</table>
</form> </body> </html>
```

Default.aspx.cs

```
using System;
using System.Collections.Generic;
using System.Linq; using
System.Web; using
System.Web.UI;
using System.Web.UI.WebControls;
namespace Experiment2
{
public partial class RegistrationPage : System.Web.UI.Page
{
protected void Page_Load(object sender, EventArgs e)
{
}
protected void Button1_Click(object sender, EventArgs e)
{
message.Text = "Hello " + username.Text + " ! ";
message.Text = message.Text + " <br/> You have successfully Registered with the following
details.";
ShowUserName.Text = username.Text;      UserEmailid.Text
= EmailID.Text;
if (RadioButton1.Checked)
{
ShowGender.Text = RadioButton1.Text;
}
else ShowGender.Text = RadioButton2.Text;      var
courses = "";
if (CheckBox1.Checked)
{
courses = CheckBox1.Text + " ";
}
if (CheckBox2.Checked)
{
courses += CheckBox2.Text + " ";
}
if (CheckBox3.Checked)
{
```

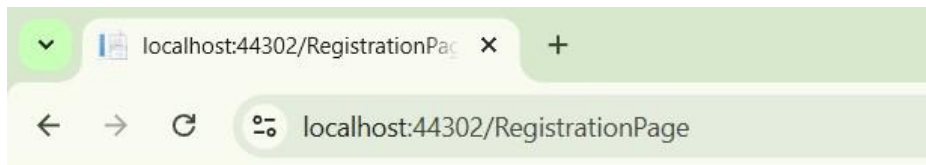
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```
courses += CheckBox3.Text;
}
Response.Write("<script>alert('Registered successfully')</script>");
ShowCourses.Text = courses;
UName.Text = "User Name";
Email.Text = "Email ID";
Gender.Text = "Gender";      Course.Text
= "Courses";      username.Text = "";
EmailID.Text = "";
RadioButton1.Checked = false;
RadioButton2.Checked = false;
CheckBox1.Checked = false;
CheckBox2.Checked = false;
CheckBox3.Checked = false;

}
}
}
```

Output:



Anchal Singh - Roll No: 119

User Name	
Email ID	
Password	
Confirm Password	
Gender	☐ Male ☐ Female
Select Courses	☐ Advance Java ☐ DotNet ☐ Mobile Computing
Register	

Hello Anchal !

You have successfully Registered with the following details.

User Name Anchal

Email ID anchal@gmail.com

Gender Female

Courses Advance Java DotNet

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3. Implement a Cross Page Postback feature to pass data between two web pages.

Webform1.aspx

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="WebForm1.aspx.cs"
Inherits="practical3.WebForm1" %>
<!DOCTYPE html>
<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
<title></title>
</head>
<body>
<form id="form1" runat="server">
<asp:Label ID="lblStudentInfo" runat="server"
Text="Anchal Singh - Roll No: 119"
Font-Bold="True"
Font-Size="Large"></asp:Label>
<br /><br />
<div>
<asp:TextBox ID="TextBox1" runat="server"></asp:TextBox>
<asp:Button ID="Button1" runat="server" Text="ClickMeUsingServerTransfer"
OnClick="Button1_Click" />
</div>
</form>
</body>
</html>
```

Webform1.aspx.cs

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;
namespace practical3
{
public partial class WebForm1 : System.Web.UI.Page
{
protected void Page_Load(object sender, EventArgs e)
{
}
protected void Button1_Click(object sender, EventArgs e)
```


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```
{  
Server.Transfer("~/WebForm2.aspx?Name= " + TextBox1.Text + " &Work=Developer");  
}  
}  
}
```

Webform2.aspx

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="WebForm2.aspx.cs"  
Inherits="practical3.WebForm2" %>
```

```
<!DOCTYPE html>  
<html xmlns="http://www.w3.org/1999/xhtml">  
<head runat="server">  
<title></title>  
</head>  
<body>  
<form id="form1" runat="server">  
<asp:Label ID="lblStudentInfo" runat="server"  
Text="Anchal Singh - Roll No: 119"  
Font-Bold="True"  
Font-Size="Large"></asp:Label>  
<br /><br />  
<div>  
<asp:Label ID="Label1" runat="server" Text="Label"></asp:Label>  
</div>  
</form>  
</body>  
</html>
```

webform2.aspx.cs

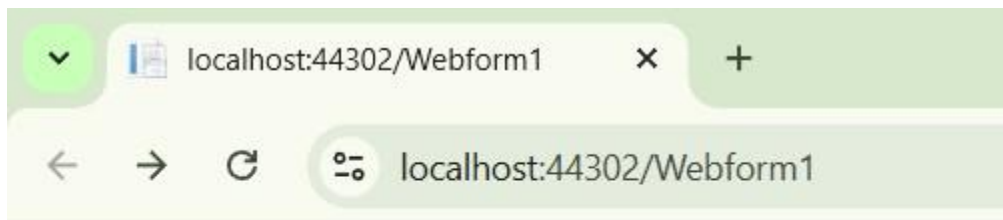
```
using System;  
using System.Collections.Generic;  
using System.Linq;  
using System.Web;  
using System.Web.UI;  
using System.Web.UI.WebControls;  
namespace practical3  
{  
public partial class WebForm2 : System.Web.UI.Page  
{  
protected void Page_Load(object sender, EventArgs e)
```

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```
{  
if (PreviousPage != null)  
{  
Label1.Text = "Welcome" + Request.QueryString["Name"] + ": .Net " +  
Request.QueryString["Work"];  
}  
  
}  
}  
}
```

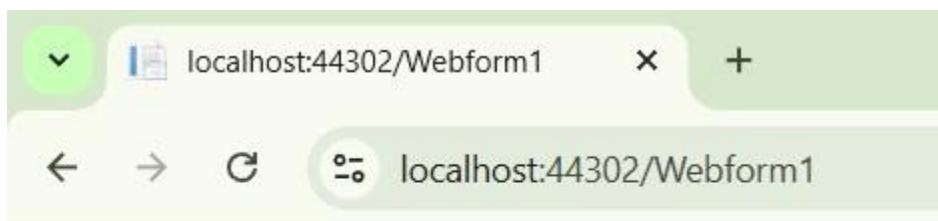
Output:



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Anchal Singh

ClickMeUsingServerTransfer



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Welcome Anchal Singh : .Net Developer

B. Postback.aspx

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```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="Postback.aspx.cs"
Inherits="PageLifeCycle.PostbackPage" %>
<!DOCTYPE html>
<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
<title>Postback Example</title>
</head>
<body>
<form id="form1" runat="server">
<asp:Label ID="lblStudentInfo" runat="server"
Text="Anchal Singh - Roll No: 119"
Font-Bold="True"
Font-Size="Large"></asp:Label>
<br /><br />
<div>
<h1>Monitoring the control for every change event:</h1>

<asp:TextBox ID="txt1" runat="server" AutoPostBack="true"
OnTextChanged="ControlChange" />
<br />

<asp:CheckBox ID="chk1" runat="server" AutoPostBack="true"
OnCheckedChanged="ControlChange" />
<br />

<asp:RadioButton ID="opt1" runat="server" GroupName="Example"
AutoPostBack="true" OnCheckedChanged="ControlChange" />
<asp:RadioButton ID="opt2" runat="server" GroupName="Example"
AutoPostBack="true" OnCheckedChanged="ControlChange" />

<h1>Required Events: </h1>
<asp:ListBox ID="listE" runat="server" Width="380px" Height="160px" /><br /><br />
</div>
</form>
</body>
</html>
```

postback.aspx.cs

```
using System; using
System.Web.UI;
using System.Web.UI.WebControls;

namespace PageLifeCycle
{
```

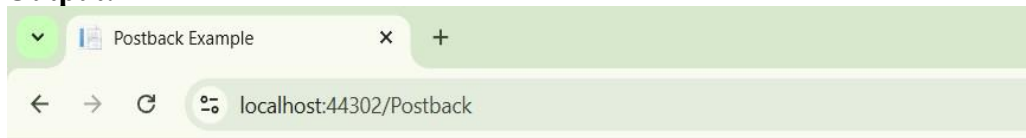
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```
public partial class PostbackPage : System.Web.UI.Page
{
protected void Page_Load(object sender, EventArgs e)
{
}

protected void ControlChange(object sender, EventArgs e)
{
string controlName = ((Control)sender).ID;
Log(controlName + " Triggered");
}
private void Log(string entry)
{
listE.Items.Add(entry);
listE.SelectedIndex = listE.Items.Count - 1;
}
}
}
```

Output:



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Monitoring the control for every change event:

☒ ☐

Required Events:

txt1 Triggered
opt11 Triggered
opt21 Triggered
chk1 Triggered

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4. Develop a web application demonstrating AutoPostBack using a dropdown list to select and display items.

Autopostback.aspx

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="AutopostBack.aspx.cs" Inherits="PageLifeCycle.WebForm1" %>
```

```
<!DOCTYPE html>
<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
<title></title>
</head>
<body>
<form id="form1" runat="server">
<div>
<asp:Label runat="server" Text="Anchal Singh 119"></asp:Label> <br /> <br />
<asp:DropDownList ID="Course" runat="server" AutoPostBack="True"
OnSelectedIndexChanged="DropDownList1_SelectedIndexChanged">
<asp:ListItem Text="MCA" Value="1"></asp:ListItem>
<asp:ListItem Text="BCA" Value="2"></asp:ListItem>
<asp:ListItem Text="BSCIT" Value="3"></asp:ListItem>
<asp:ListItem Text="BSCCS" Value="4"></asp:ListItem>
<asp:ListItem Text="MSCIT" Value="5"></asp:ListItem>
</asp:DropDownList>
</div>
</form>
</body>
</html>
```

Autopostback.cs

```
using System;
using System.Collections.Generic;
using System.Linq; using
System.Web; using
System.Web.UI; using
System.Web.UI.WebControls;
namespace PageLifeCycle
{
public partial class WebForm1 : System.Web.UI.Page
{
```

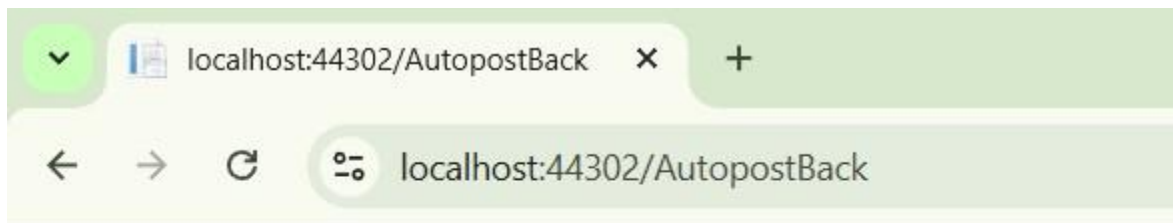
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```
protected void Page_Load(object sender, EventArgs e)
{
}
protected void DropDownList1_SelectedIndexChanged(object sender, EventArgs e)
{
    string message = Course.SelectedItem.Text + " - " + Course.SelectedItem.Value;
    ClientScript.RegisterStartupScript(this.GetType(), "alert", "alert('" + message + "');", true);
}

}
}
```

Output:



Anchal Singh 119

MCA ▼



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5. Create a simple calculator using server-side controls and operators in ASP.NET.

CalculatorForm.aspx

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="CalculatorForm.aspx.cs"
Inherits="PageLifeCycle.CalculatorForm" %>
<!DOCTYPE html>
<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
<title>Calculator</title>
<style>
.calculator { border-collapse: collapse; }
.calc_btn { width: 50px; height: 50px; font-size: 18px; }
.calc_result { width: 200px; height: 40px; font-size: 20px; text-align: right; }
</style>
</head>
<body>
<form id="form1" runat="server">
<asp:Label runat="server" Text="Anchal Singh 119"></asp:Label>
<div class="calculator-holder">
<table class="calculator">
<tr>
<td colspan="4">
<input type="text" readonly="readonly" runat="server" id="calc_result" class="calc_result" />
</td>
</tr><tr>
<td><asp:Button ID="ButtonCE" runat="server" CssClass="calc_btn" Text="CE"
OnClick="ButtonCE_Click" /></td>
<td><asp:Button ID="ButtonReturn" runat="server" CssClass="calc_btn"
Text="&larr;" OnClick="ButtonReturn_Click" /></td>
<td><asp:Button ID="ButtonPercentage" runat="server" CssClass="calc_btn"
Text="%" OnClick="ButtonPercentage_Click" /></td>
<td><asp:Button ID="ButtonPlus" runat="server" CssClass="calc_btn" Text="+"
OnClick="ButtonPlus_Click" /></td>
</tr><tr>
<td><asp:Button ID="Button7" runat="server" CssClass="calc_btn" Text="7"
OnClick="Button7_Click" /></td>
<td><asp:Button ID="Button8" runat="server" CssClass="calc_btn" Text="8"
OnClick="Button8_Click" /></td>
<td><asp:Button ID="Button9" runat="server" CssClass="calc_btn" Text="9"
OnClick="Button9_Click" /></td>
<td><asp:Button ID="ButtonMinus" runat="server" CssClass="calc_btn" Text="-"
OnClick="ButtonMinus_Click" /></td>
```

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```
</tr><tr>
<td><asp:Button ID="Button4" runat="server" CssClass="calc_btn" Text="4"
OnClick="Button4_Click" /></td>
<td><asp:Button ID="Button5" runat="server" CssClass="calc_btn" Text="5"
OnClick="Button5_Click" /></td>
<td><asp:Button ID="Button6" runat="server" CssClass="calc_btn" Text="6"
OnClick="Button6_Click" /></td>
<td><asp:Button ID="ButtonMultiply" runat="server" CssClass="calc_btn"
Text="x" OnClick="ButtonMultiply_Click" /></td>
</tr><tr>
<td><asp:Button ID="Button1" runat="server" CssClass="calc_btn" Text="1"
OnClick="Button1_Click" /></td>
<td><asp:Button ID="Button2" runat="server" CssClass="calc_btn" Text="2"
OnClick="Button2_Click" /></td>
<td><asp:Button ID="Button3" runat="server" CssClass="calc_btn" Text="3"
OnClick="Button3_Click"/></td>
<td><asp:Button ID="ButtonDivide" runat="server" CssClass="calc_btn"
Text="&divide;" OnClick="ButtonDivide_Click" /></td>
</tr><tr>
<td><asp:Button ID="Button0" runat="server" CssClass="calc_btn" Text="0"
OnClick="Button0_Click" /></td>
<td><asp:Button ID="ButtonPlusMinus" runat="server" CssClass="calc_btn"
Text="&plusmn;" OnClick="ButtonPlusMinus_Click"/></td>
<td><asp:Button ID="ButtonComa" runat="server" CssClass="calc_btn" Text=","
OnClick="ButtonComa_Click" /></td>
<td><asp:Button ID="ButtonEquals" runat="server" CssClass="calc_btn" Text="="
OnClick="ButtonEquals_Click"/></td>
</tr></table></div></form></body></html>
```

CalculatorForm.aspx.cs

```
using System;
using System.Web.UI;
namespace PageLifeCycle
{
    public partial class CalculatorForm : Page
    {
        Calculator _calculator;
        protected void Page_Load(object sender, EventArgs e)
        {
            _calculator = new Calculator();
        }
        // Number buttons
        protected void Button0_Click(object sender, EventArgs e) => calc_result.Value += "0";
        protected void Button1_Click(object sender, EventArgs e) => calc_result.Value += "1";
```


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```
protected void Button2_Click(object sender, EventArgs e) => calc_result.Value += "2";
protected void Button3_Click(object sender, EventArgs e) => calc_result.Value += "3";
protected void Button4_Click(object sender, EventArgs e) => calc_result.Value += "4";
protected void Button5_Click(object sender, EventArgs e) => calc_result.Value += "5";
protected void Button6_Click(object sender, EventArgs e) => calc_result.Value += "6";
protected void Button7_Click(object sender, EventArgs e) => calc_result.Value += "7";
protected void Button8_Click(object sender, EventArgs e) => calc_result.Value += "8";
protected void Button9_Click(object sender, EventArgs e) => calc_result.Value += "9";

protected void ButtonComa_Click(object sender, EventArgs e) => calc_result.Value += ",";
protected void ButtonPlusMinus_Click(object sender, EventArgs e)
{
    if (string.IsNullOrEmpty(calc_result.Value))
        Response.Write("<script>alert('No Value is given.')</script>");
    else
        calc_result.Value = "-" + calc_result.Value;
}
// Operations
protected void ButtonPlus_Click(object sender, EventArgs e) => SetOperation("Addition");
protected void ButtonMinus_Click(object sender, EventArgs e) => SetOperation("Subtraction");
protected void ButtonMultiply_Click(object sender, EventArgs e) =>
    SetOperation("Multiplication");
protected void ButtonDivide_Click(object sender, EventArgs e) => SetOperation("Division");
protected void ButtonPercentage_Click(object sender, EventArgs e) =>
    SetOperation("Percentage");

private void SetOperation(string operation)
{
    if (string.IsNullOrEmpty(calc_result.Value))
        Response.Write("<script>alert('No Value is given.')</script>");
    else
    {
        ViewState["Value1"] = calc_result.Value;
        ViewState["Operation"]
        = operation;
        calc_result.Value = string.Empty;
    }
}
protected void ButtonCE_Click(object sender, EventArgs e)
{
    ViewState["Operation"] = null;
    ViewState["Value1"] = null;
    calc_result.Value = string.Empty;
}
protected void ButtonReturn_Click(object sender, EventArgs e)
{
    if (!string.IsNullOrEmpty(calc_result.Value))
        calc_result.Value = calc_result.Value.Substring(0, calc_result.Value.Length - 1);
}
```

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```
}
protected void ButtonEquals_Click(object sender, EventArgs e)
{
    if (string.IsNullOrEmpty(calc_result.Value))
    {
        Response.Write("<script>alert('No Value is given.')</script>");
        return;
    }
    ViewState["Value2"] = calc_result.Value;
    calc_result.Value = string.Empty;
    try
    {
        string op = (string)ViewState["Operation"];
        int val1 = Convert.ToInt32(ViewState["Value1"]);
        int val2 = Convert.ToInt32(ViewState["Value2"]);
        switch (op)
        {
            case "Addition":
                calc_result.Value = _calculator.Add(val1, val2).ToString();
                break;
            case "Subtraction":
                calc_result.Value = _calculator.Subtract(val1, val2).ToString();
                break;
            case "Multiplication":
                calc_result.Value = _calculator.Multiply(val1, val2).ToString();
                break;
            case "Division":
                calc_result.Value = _calculator.Divide(val1, val2).ToString();
                break;
            case "Percentage":
                calc_result.Value = _calculator.Percentage(val1, val2);
                break;
            default:
                Response.Write("<script>alert('No Operation was recorded.')</script>");
                break;
        }
    }
    catch (FormatException)
    {
        Response.Write("<script>alert('Bad Input Format.')</script>");
    }
}
```

Calculator.cs

Project-> right click -> Add class

using System;

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namespace PageLifeCycle

{

public class Calculator

{

public int Add(int value1, int value2) => value1 + value2;

public int Subtract(int value1, int value2) => value1 - value2;

public int Multiply(int value1, int value2) => value1 * value2;

public double Divide(int value1, int value2) => value1 / (double)value2;

public string Percentage(int value1, int value2)

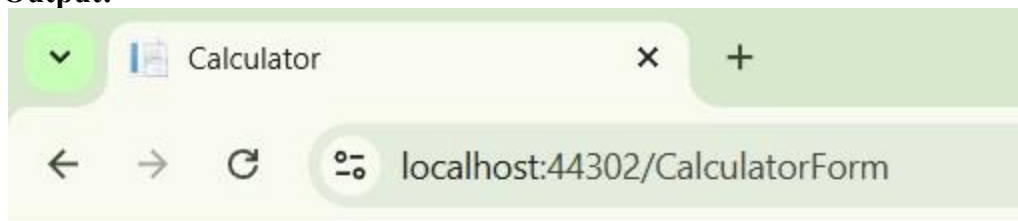
{

double result = (value1 * 100.0) / value2; return

result + "%";

}}}

Output:



Anchal Singh 119

6			
CE	←	%	+
7	8	9	-
4	5	6	x
1	2	3	÷
0	±	,	=

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Module 2: Creating a User Interface

1. Design a registration form with validation controls for mandatory fields, email format, and passwords.

Webform.aspx

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="Webform.aspx.cs"
Inherits="Practical6.WebForm1" %>
<!DOCTYPE html>
<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
<title></title>
</head>
<body>
<form id="form1" runat="server">
<div>
<h2>Registration Form</h2>
<label>Enter Full Name:</label>
<asp:TextBox ID="txtName" runat="server"></asp:TextBox>
<asp:RequiredFieldValidator ID="rfvName" runat="server"
ControlToValidate="txtName" ErrorMessage="Full Name is required."
CssClass="error" Display="Dynamic" />
<br /><br />
<label>Enter Address:</label>
<asp:TextBox ID="txtAddress" runat="server" TextMode="MultiLine"></asp:TextBox>
<asp:RequiredFieldValidator ID="rfvAddress" runat="server"
ControlToValidate="txtAddress" ErrorMessage="Address is required."
CssClass="error" Display="Dynamic" />
<br /><br />
<label>Enter Contact Number:</label>
<asp:TextBox ID="txtContact" runat="server"></asp:TextBox>
<asp:RequiredFieldValidator ID="rfvContact" runat="server"
ControlToValidate="txtContact" ErrorMessage="Contact Number is required."
CssClass="error" Display="Dynamic" />
<asp:RegularExpressionValidator ID="revContact" runat="server"
ControlToValidate="txtContact"
ValidationExpression="\d{10}$"
ErrorMessage="Enter a valid 10-digit number."
CssClass="error" Display="Dynamic" />
<br /><br />
<label>Enter Password:</label>
<asp:TextBox ID="txtPassword" runat="server" TextMode="Password"></asp:TextBox>
<asp:RequiredFieldValidator ID="rfvPassword" runat="server"
ControlToValidate="txtPassword" ErrorMessage="Password is required."
CssClass="error" Display="Dynamic" />
```

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```
<br /><br />
<label>Confirm Password:</label>
<asp:TextBox ID="txtConfirmPassword" runat="server"
TextMode="Password"></asp:TextBox>
<asp:RequiredFieldValidator ID="rfvConfirmPassword" runat="server"
ControlToValidate="txtConfirmPassword" ErrorMessage="Confirm Password is required."
CssClass="error" Display="Dynamic" />
<asp:CompareValidator ID="cvPassword" runat="server"
ControlToValidate="txtConfirmPassword" ControlToCompare="txtPassword"
ErrorMessage="Passwords do not match."
CssClass="error" Display="Dynamic" />
<br /><br />

<asp:Button ID="btnRegister" runat="server" Text="Register"
OnClick="btnRegister_Click" />
<asp:Label ID="lblMessage" runat="server" ForeColor="Green"></asp:Label>
</div>
</form>
</body>
</html>
```

webform.aspx.cs

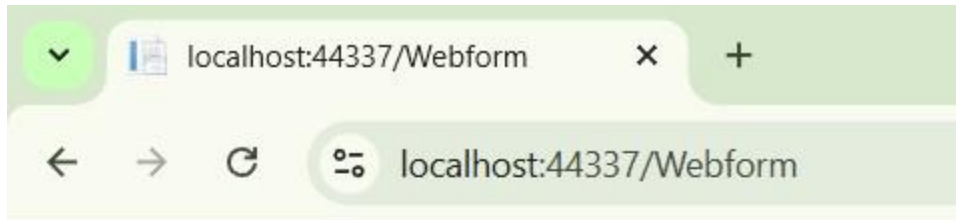
```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;
namespace Practical6
{
    public partial class WebForm1 : System.Web.UI.Page
    {
        protected void Page_Load(object sender, EventArgs e)
        {
            UnobtrusiveValidationMode = UnobtrusiveValidationMode.None;
        }
        protected void btnRegister_Click(object sender, EventArgs e)
        {
            if (Page.IsValid) // Ensure all validations are passed
            {
                lblMessage.Text = "Registration Successful!";
            }
        }
    }
}
```

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}

Output:



Registration Form

Enter Full Name:

Enter Address:

Enter Contact Number:

Enter Password:

Confirm Password:

Registration Successful!

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2. Design an ASP.NET application to display random advertisements using AdRotator control.

Webform2.aspx

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="Webform2.aspx.cs"
Inherits="Practical6.WebForm2" %>
<!DOCTYPE html>
<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
<title></title>
</head>
<body>
<form id="form1" runat="server">
<label>Anchal Singh 119</label>
<div>
<asp:AdRotator ID="AdRotator1" runat="server"
AdvertisementFile="~/XMLFile1.xml" Height="380px"
OnAdCreated="AdRotator1_AdCreated" Width="780px" />
</div></form></body></html>
```

Webform2.aspx.cs

```
using System;
using System.Web.UI.WebControls;
namespace Practical6
{
public partial class WebForm2 :
System.Web.UI.Page
{
protected void Page_Load(object sender, EventArgs e)
{}
protected void AdRotator1_AdCreated(object sender, AdCreatedEventArgs e)
{}}}
```

Xml file'

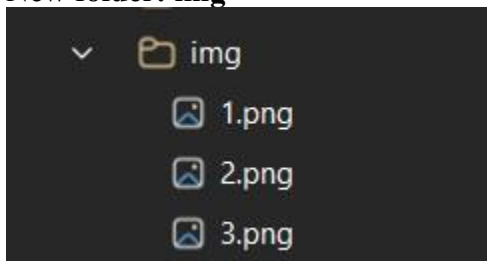
```
<?xml version="1.0" encoding="utf-8" ?>
<Advertisements>
<Ad>
<ImageUrl>img/1.png</ImageUrl>
<NavigateUrl>https://google.com</NavigateUrl>
<AlternateText>Meera Academy</AlternateText>
<Impressions>50</Impressions>
<Keyword>Meera</Keyword>
```

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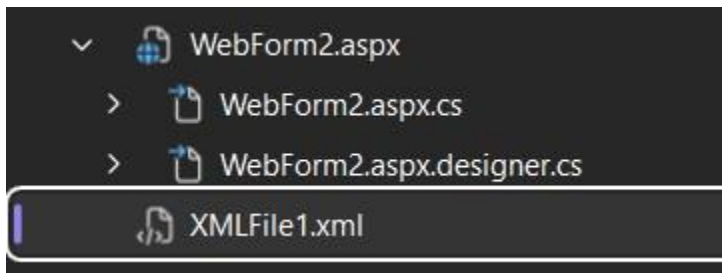
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```
</Ad>
<Ad>
<ImageUrl>img/2.png</ImageUrl>
<NavigateUrl>http://zomato.com</NavigateUrl>
<AlternateText>Meerta Aademy</AlternateText>
<Impressions>100</Impressions>
<Keyword>Academy</Keyword>
</Ad>
<Ad>
<ImageUrl>img/3.png</ImageUrl>
<NavigateUrl>https://swiggy.com</NavigateUrl>
<AlternateText>Meerta
Aademy</AlternateText>
<Impressions>50</Impressions>
<Keyword>Academy</Keyword>
</Ad>
</Advertisements>
```

New folder: img

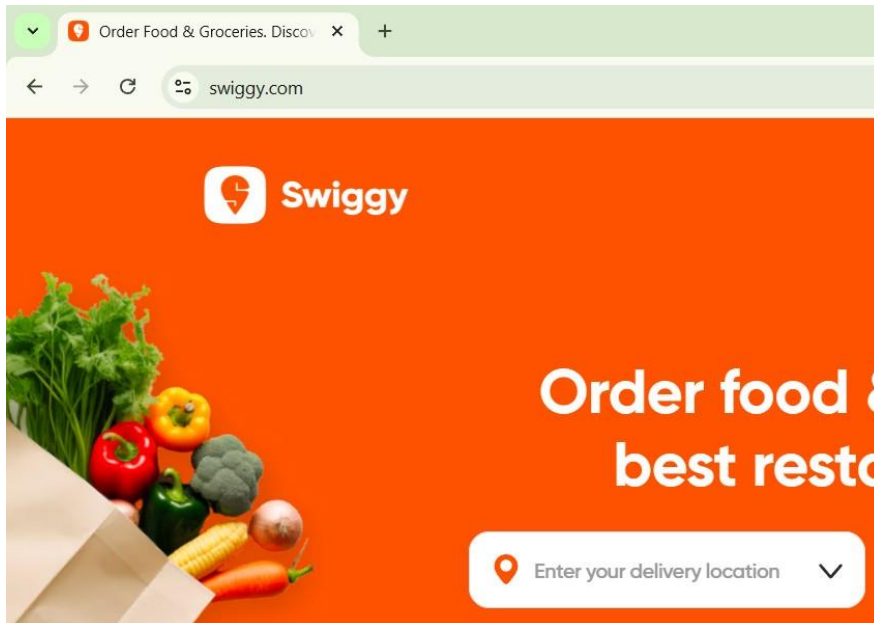
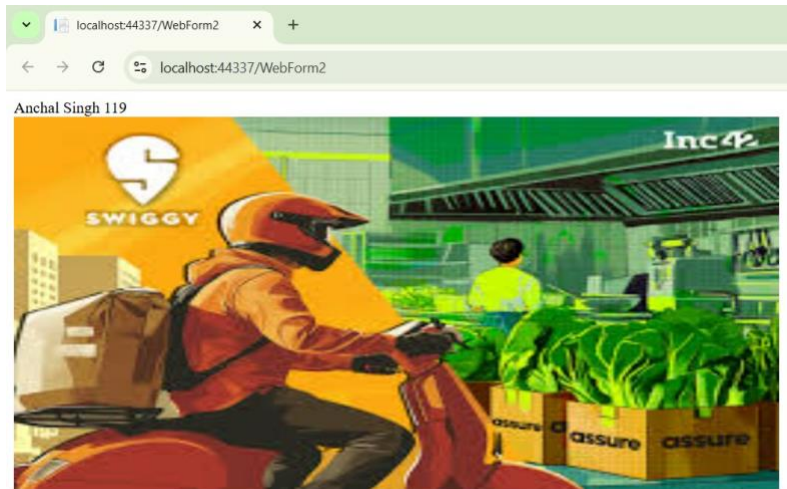


New Item -> Xmlfile.xml



Output:

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Practical 8

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3. Create a web page with an interactive calendar control to select a date and display it in a label. Display the current month's calendar with an image in the cell and a message.

Webform.aspx

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="WebForm.aspx.cs"
Inherits="Practical_8.WebForm" %>
<!DOCTYPE html>
<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
<title></title></head><body>
<form id="form1" runat="server">
<div>
<p style="text-align: center">
<b>
<asp:Label ID="Label1" runat="server" Font-Bold="True" Font-Names="Arial Black"
Font-Size="Medium"
ForeColor="#0066FF">Anchal Singh 119</asp:Label><br /></b>
</p>
<p style="text-align: center">
<b>
<asp:Label ID="Label2" runat="server" Font-Bold="True" Font-Names="Arial Black"
Font-Size="Medium"
ForeColor="#0066FF">Indian List of Holidays 2025</asp:Label><br /></b>
</p>
<asp:Calendar ID="Calendar1" runat="server" BackColor="#FFFFCC"
BorderColor="#FFCC66"
BorderWidth="1px" DayNameFormat="Shortest" Font-Names="Verdana" Font-Size="8pt"
ForeColor="#663399" ShowGridLines="True" OnDayRender="Calendar1_DayRender"
OnSelectionChanged="Calendar1_SelectionChanged"
OnVisibleMonthChanged="Calendar1_VisibleMonthChanged">
<SelectedDayStyle BackColor="#CCCCFF" Font-Bold="True" />
<SelectorStyle BackColor="#FFCC66" />
<TodayDayStyle BackColor="#FFCC66" ForeColor="White" />
<OtherMonthDayStyle ForeColor="#CC9966" />
<NextPrevStyle Font-Size="9pt" ForeColor="#FFFFCC" />
<DayHeaderStyle BackColor="#FFCC66" Font-Bold="True" Height="1px" />
<TitleStyle BackColor="#990000" Font-Bold="True" Font-Size="9pt"
ForeColor="#FFFFCC" />
</asp:Calendar>
<br /><b></b>
<asp:Label ID="LabelAction" runat="server"></asp:Label><br />
</b></div></form></body></html>
```

Webform.aspx.cs

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```
using System;
using System.Collections;
using System.Collections.Generic;
using System.Linq; using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;
namespace Practical_8
{
    public partial class WebForm : System.Web.UI.Page
    {
        Hashtable HolidayList;
        protected void Page_Load(object sender, EventArgs e)
        {
            HolidayList = Getholiday();
            Calendar1.Caption = "Calender ";
            Calendar1.FirstDayOfWeek = FirstDayOfWeek.Sunday;
            Calendar1.NextPrevFormat = NextPrevFormat.ShortMonth;
            Calendar1.TitleFormat = TitleFormat.Month;
            Calendar1.ShowGridLines = true;
            Calendar1.DayStyle.Height = new Unit(50);
            Calendar1.DayStyle.Width = new Unit(150);
            Calendar1.DayStyle.HorizontalAlign = HorizontalAlign.Center;
            Calendar1.DayStyle.VerticalAlign = VerticalAlign.Middle;
            Calendar1.OtherMonthDayStyle.BackColor = System.Drawing.Color.AliceBlue;
        }
        private Hashtable Getholiday()
        {
            Hashtable holiday = new Hashtable();
            holiday["1/1/2025"] = "New Year";
            holiday["1/5/2025"] = "Guru Govind Singh Jayanti";
            holiday["1/8/2025"] = "Muharram (Al Hijra)";
            holiday["1/14/2025"] = "Pongal";
            holiday["1/26/2025"] = "Republic Day";
            holiday["2/23/2025"] = "Maha Shivaratri";
            holiday["3/10/2025"] = "Milad un Nabi (Birthday of the Prophet)";
            holiday["3/21/2025"] = "Holi";
            holiday["3/21/2025"] = "Telugu New Year";
            holiday["4/3/2025"] = "Ram Navami";
            holiday["4/7/2025"] = "Mahavir Jayanti";
            holiday["4/10/2025"] = "Good Friday";
            holiday["4/12/2025"] = "Easter";
            holiday["4/14/2025"] = "Tamil New Year and Dr Ambedkar Birth Day";
            holiday["5/1/2025"] = "May Day";
            holiday["5/9/2025"] = "Buddha Jayanti and Buddha Purnima";
            holiday["6/24/2025"] = "Rath Yatra";
            holiday["8/13/2025"] = "Krishna Jayanthi";
            holiday["8/14/2025"] = "Janmashtami";
            holiday["8/15/2025"] = "Independence Day";
            holiday["8/19/2025"] = "Parsi New Year";
            holiday["8/23/2025"] = "Vinayaka Chaturthi";
            holiday["9/2/2025"] = "Onam";
            holiday["9/5/2025"] = "Teachers Day";
            holiday["9/21/2025"] = "Ramzan";
            holiday["9/27/2025"] = "Ayutha Pooja";
            holiday["9/28/2025"] = "Vijaya Dasami (Dusherra)";
            holiday["10/2/2025"] = "Gandhi Jayanti";
            holiday["10/17/2025"] = "Diwali &
```

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```
Govardhan Puja";        holiday["10/19/2025"] = "Bhaidooj";        holiday["11/2/2025"] =
"Guru Nanak Jayanti";        holiday["11/14/2025"] = "Children's Day";
holiday["11/28/2025"] = "Bakrid";        holiday["12/25/2025"] = "Christmas";
holiday["12/28/2025"] = "Muharram";
return holiday;
}
protected void Calendar1_SelectionChanged(object sender, EventArgs e)
{
    LabelAction.Text = "Date changed to :" + Calendar1.SelectedDate.ToShortDateString();
}
protected void Calendar1_VisibleMonthChanged(object sender, MonthChangedEventArgs e)
{
    LabelAction.Text = "Month changed to :" + e.NewDate.ToShortDateString();
}
protected void Calendar1_DayRender(object sender, DayRenderEventArgs e)
{
    if (HolidayList[e.Day.Date.ToShortDateString()] != null)
    {
        Literal literal1 = new Literal();        literal1.Text = "<br/>";
        e.Cell.Controls.Add(literal1);        Label label1 = new Label();
        label1.Text = (string)HolidayList[e.Day.Date.ToShortDateString()];        label1.Font.Size
        = new FontUnit(FontSize.Small);        e.Cell.Controls.Add(label1);
    }
}
```

Output:

Calendar

Feb	Su	Mo	Tu	We	Th	Fr	Sa	Apr
	22	23	24	25	26	27	28	
	1	2	3	4	5	6	7	
	8	9	10	11	12	13	14	
	15	16	17	18	19	20	21	
	22	23	24	25	26	27	28	
	29	30	31	1	2	3	4	

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4. Implement a navigation menu to link multiple pages of your web application.

Webform.aspx

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="WebForm1.aspx.cs"
Inherits="Practical_9.WebForm1" %>

<!DOCTYPE html>
<html>
<head runat="server">
<title>TreeView Navigation</title>
</head><body>
<form id="form1" runat="server">
<label>Anchal Singh 119</label>
<div>
<asp:TreeView ID="TreeView1" runat="server"
DataSourceID="SiteMapDataSource1"
ImageSet="Simple"
NodeIndent="10">
<HoverNodeStyle Font-Underline="True" ForeColor="Red" />
</asp:TreeView>
<asp:SiteMapDataSource ID="SiteMapDataSource1" runat="server"
ShowStartingNode="false" />
</div></form></body></html>
```

Menuexample.aspx

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="Menuexample.aspx.cs"
Inherits="Practical_9.WebForm2" %>
<!DOCTYPE html>
<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
<title></title>
<style type="text/css">
body {
background-color: mediumaquamarine;
font-family: Arial;
font-size: 10pt;
color: #444;
}
.ParentMenu, .ParentMenu:hover {
width: 100px;
background-color: #fff;
color: #333;
text-align: center;
height: 30px;
```

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```
line-height: 30px;
margin-right: 5px;
}
.ParentMenu:hover {
background-color: #ccc;
}
.ChildMenu, .ChildMenu:hover {
width: 110px;
background-color: #fff;
color: #333;
text-align: center;
height: 30px;
line-height: 30px;
margin-top: 5px;
}
.ChildMenu:hover {
background-color: #ccc;
}
.selected, .selected:hover {
background-color: #A6A6A6 !important;
color: #fff;
}
.level2 {
background-color: #fff;
}
</style></head>
<body>
<form id="form1" runat="server">
<div>
<asp:Menu ID="Menu1" runat="server" Orientation="Horizontal">
<LevelMenuItemStyles>
<asp:MenuItemStyle CssClass="ParentMenu" />
<asp:MenuItemStyle CssClass="ChildMenu" />
<asp:MenuItemStyle CssClass="ChildMenu" />
</LevelMenuItemStyles>
<StaticSelectedStyle CssClass="selected" />
</asp:Menu>
</div>
</form></body></html>
```

Menuexample.aspx.cs

```
using System;
using System.Collections.Generic;
using System.Data; using System.IO;
using System.Linq; using System.Web;
```

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```
using System.Web.UI;
using System.Web.UI.WebControls;
namespace Practical_9
{
    public partial class WebForm2 : System.Web.UI.Page
    {
        protected void Page_Load(object sender, EventArgs e)
        {
            if (!IsPostBack)
            {
                DataTable dt = this.BindMenuData(0);
                DynamicMenuControl.Population(dt, 0, null);
            }
        }
        protected DataTable BindMenuData(int parentmenuId)
        {
            DataSet ds = new DataSet();
            DataTable dt;
            DataRow dr;
            DataColumn menu;
            DataColumn pMenu;
            DataColumn title;
            DataColumn description;
            DataColumn URL; dt = new DataTable();
            menu = new DataColumn("MenuId", Type.GetType("System.Int32"));
            pMenu = new DataColumn("ParentId", Type.GetType("System.Int32"));
            title = new DataColumn("Title", Type.GetType("System.String"));
            description = new DataColumn("Description", Type.GetType("System.String"));
            URL = new DataColumn("URL", Type.GetType("System.String"));
            dt.Columns.Add(menu);
            dt.Columns.Add(pMenu);
            dt.Columns.Add(title);
            dt.Columns.Add(description);
            dt.Columns.Add(URL);
            dr = dt.NewRow();
            dr["MenuId"] = 1;
            dr["ParentId"] = 0;
            dr["Title"] = "Home";
            dr["Description"] = "";
            dr["URL"] = "~/Home.aspx";
            dt.Rows.Add(dr);
            dr = dt.NewRow();
            dr["MenuId"] = 2;
            dr["ParentId"] = 0;
            dr["Title"] = "Customer Service";
            dr["Description"] = "Customer Service";
            dr["URL"] = "~/Customer.aspx";
```

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```
dt.Rows.Add(dr);
dr = dt.NewRow();
dr["MenuId"] = 3;
dr["ParentId"] = 0;
dr["Title"] = "About";
dr["Description"] = "About us page";
dr["URL"] = "~/AboutUs.aspx";
dt.Rows.Add(dr);
dr = dt.NewRow();
dr["MenuId"] = 4;
dr["ParentId"] = 0;
dr["Title"] = "Contact Us";
dr["Description"] = "Contact Us page";
dr["URL"] = "~/Contact.aspx";
dt.Rows.Add(dr);
dr = dt.NewRow();
dr["MenuId"] = 5;
dr["ParentId"] = 0;
dr["Title"] = "Testmonial";
dr["Description"] = "Testimonial page";
dr["URL"] = "~/Testimonial.aspx";
dt.Rows.Add(dr);
dr = dt.NewRow();
dr["MenuId"] = 6;
dr["ParentId"] = 2;
dr["Title"] = "Consulting";
dr["Description"] = "Consulting page";
dr["URL"] = "~/Consult.aspx";
dt.Rows.Add(dr);
dr = dt.NewRow();
dr["MenuId"] = 7;
dr["ParentId"] = 2;
dr["Title"] = "Outsourcing";
dr["Description"] = "Outsourcing page";
dr["URL"] = "~/Outsource.aspx";
dt.Rows.Add(dr);
dr = dt.NewRow();
dr["MenuId"] = 8;
dr["ParentId"] = 7;
dr["Title"] = "Domestic";
dr["Description"] = "Domestic outsourcing page";
dr["URL"] = "~/Domestic.aspx";
dt.Rows.Add(dr);
dr = dt.NewRow();
dr["MenuId"] = 9;
dr["ParentId"] = 7;
```


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```
dr["Title"] = "International";
dr["Description"] = "International outsourcing page";
dr["URL"] = "~/International.aspx";
dt.Rows.Add(dr);
ds.Tables.Add(dt);
var dv = ds.Tables[0].DefaultView;
dv.RowFilter = "ParentId=" + parentmenuId + "";
DataSet ds1 = new DataSet();          var newdt = dv.ToTable();
return newdt;
}
protected void DynamicMenuControlPopulation(DataTable dt, int parentMenuId, MenuItem
parentMenuItem)
{
string currentPage = Path.GetFileName(Request.Url.AbsolutePath); foreach (DataRow row in
dt.Rows) {
MenuItem menuItem = new MenuItem
{
Value = row["MenuId"].ToString(),
Text = row["Title"].ToString(),
NavigateUrl = row["URL"].ToString(),
Selected = row["URL"].ToString().EndsWith(currentPage,
StringComparison.CurrentCultureIgnoreCase)
};
if (parentMenuId == 0)
{
Menu1.Items.Add(menuItem);
DataTable dtChild = this.BindMenuData(int.Parse(menuItem.Value));
DynamicMenuControlPopulation(dtChild, int.Parse(menuItem.Value), menuItem);
}
else
{
parentMenuItem.ChildItems.Add(menuItem);
DataTable dtChild = this.BindMenuData(int.Parse(menuItem.Value));
DynamicMenuControlPopulation(dtChild, int.Parse(menuItem.Value), menuItem);
}}}}}
```

Treeviewhover.aspx

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="Treeviewhover.aspx.cs"
Inherits="Practical_9.WebForm3" %>
<!DOCTYPE html>
<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
<title></title></head><body>
<form id="form1" runat="server">
<div style="font-family: Arial">
```

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```
<asp:TreeView runat="server" ID="TreeView1" ImageSet="Arrows">
<HoverNodeStyle Font-Bold="True" Font-Underline="True" ForeColor="#5555DD"
/>
<LeafNodeStyle BackColor="#FF6600" />
<Nodes>
<asp:TreeNode Text="Home" NavigateUrl="~/Default.aspx" Target="_blank" />
<asp:TreeNode Text="Employee" NavigateUrl="~/Employee.aspx" Target="_blank">
<asp:TreeNode Text="Upload Resume" NavigateUrl="~/Upload_Resume.aspx"
Target="_blank" />
<asp:TreeNode Text="Edit Resume" NavigateUrl="~/Edit_Resume.aspx" Target="_blank" />
<asp:TreeNode Text="View Resume" NavigateUrl="~/View_Resume.aspx"
Target="_blank"/>
</asp:TreeNode>
<asp:TreeNode Text="Employer" NavigateUrl="~/Employer.aspx" Target="_blank">
<asp:TreeNode Text="Upload Job" NavigateUrl="~/Upload_Job.aspx" Target="_blank" />
<asp:TreeNode Text="Edit Job" NavigateUrl="~/Edit_Job.aspx" Target="_blank"
/>
<asp:TreeNode Text="View Job" NavigateUrl="~/View_Job.aspx" Target="_blank" />
</asp:TreeNode>
<asp:TreeNode Text="Admin" NavigateUrl="~/Admin.aspx" Target="_blank">
<asp:TreeNode Text="Add User" NavigateUrl="~/Add_User.aspx" Target="_blank" />
<asp:TreeNode Text="Edit User" NavigateUrl="~/Edit_Use.aspx" Target="_blank" />
<asp:TreeNode Text="View User" NavigateUrl="~/View_User.aspx" Target="_blank" />
</asp:TreeNode>
</Nodes>
<NodeStyle Font-Names="Verdana" Font-Size="8pt" ForeColor="Black"
HorizontalPadding="5px" NodeSpacing="0px" VerticalPadding="0px" />
<ParentNodeStyle Font-Bold="False" />
<RootNodeStyle BackColor="#009900" />
<SelectedNodeStyle Font-Underline="True" ForeColor="#5555DD"
HorizontalPadding="0px" VerticalPadding="0px" />
</asp:TreeView>
</div></form></body></html>
```

Treeviewhover.aspx.cs

web.sitemap

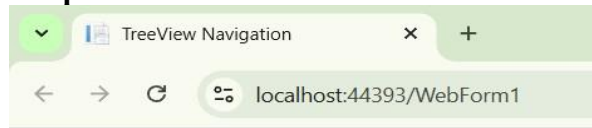
```
<?xml version="1.0" encoding="utf-8" ?>
<siteMap xmlns="http://schemas.microsoft.com/AspNet/SiteMap-File-1.0">
<siteMapNode url="" title="Home" description="">
<siteMapNode url="Home.aspx" title="Home" description="Home Page" />
<siteMapNode url="javascript:;" title="Services" description="Services"> <siteMapNode
url="TreeViewHover.aspx" title="Consulting" description="TreeViewHover" />
<siteMapNode url="TreeViewSimple.aspx" title="Outsourcing"
description="TreeViewSimple" />
</siteMapNode>
```

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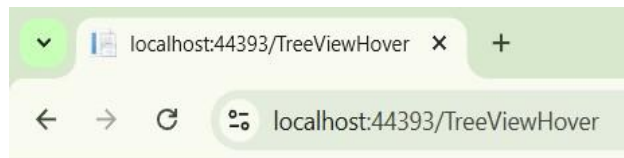
```
<siteMapNode url="MenuExample.aspx" title="About" description="MenuExample" />
<siteMapNode url="Contact.aspx" title="Contact" description="Contact Us Page" />
</siteMapNode>
</siteMap>
```

Output:



Anchal Singh 119

- Home
- Services
 - Consulting
 - Outsourcing
- About
- Contact



- ▷ Home
 - ▼ Employee
 - ▷ Upload Resume
 - ▷ Edit Resume
 - ▷ View Resume
 - ▼ Employer
 - ▷ Upload Job
 - ▷ Edit Job
 - ▷ View Job
 - ▼ Admin
 - ▷ Add User
 - ▷ Edit User
 - ▷ View User



Practical 10

Roll No: 119

5. Develop a file upload application that saves the uploaded file and displays its name and size.

Defaults.aspx

[illegible]

Default.aspx.cs

```
using System;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;
namespace Practical_9
{
    public partial class WebForm1 : System.Web.UI.Page
    {
        protected void Page_Load(object sender, EventArgs e)
        {

```

Name: Anchal Singh

Roll No: 119

```
}
string filename = string.Empty;
protected void Button_Click(object sender, EventArgs e)
{
    if ((FileUpload.PostedFile != null) && (FileUpload.PostedFile.ContentLength > 0))
    {
        filename = System.IO.Path.GetFileName(FileUpload.PostedFile.FileName);
        string SaveLocation = Server.MapPath("upload") + "\\" + filename;
        try
        {
            FileUpload.PostedFile.SaveAs(SaveLocation);
            UploadStatus.Text = "The file has been uploaded successfully.";
            lblFilename.Text = "upload/" + FileUpload.FileName;
        }
        catch (Exception ex)
        {
            UploadStatus.Text = "Error: " + ex.Message;
        }
        else
        {
            UploadStatus.Text = "Please select a file to upload.";
        }
    }
}

protected void downloadButton_Click(object sender, EventArgs e)
{
    if (!string.IsNullOrEmpty(lblFilename.Text))
    {
        string filePath = lblFilename.Text;
        string fileExtension =
        System.IO.Path.GetExtension(filePath).ToLower();
        switch (fileExtension)
        {
            case ".txt":
                Response.ContentType = "application/txt";
                break;
            case ".pdf":
                Response.ContentType = "application/pdf";
                break;
            case ".docx":
                Response.ContentType =
                "application/vnd.openxmlformats-officedocument.wordprocessingml.document";
                break;
            default:
                Response.ContentType = "image/jpeg";
                break;
        }
        Response.AddHeader("Content-Disposition", "attachment;filename=\"" +
        System.IO.Path.GetFileName(filePath) + "\"");
        Response.TransmitFile(Server.MapPath(filePath));
        Response.End();
    }
}
```

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Roll No: 119

Output:



Anchal Singh 119

Browse to Upload File

No file chosen

The file has been uploaded successfully.

Click on the button to download a file

upload/databricks.png

Name: Anchal Singh

Roll No: 119

Aim: Create a Master Page for a website and add different content pages for Home, About, and Contact Us.

Master_Page.master

```
<%@ Master Language="C#" AutoEventWireup="true"
CodeBehind="Master_Page.Master.cs" Inherits="Practical_11.Site1" %>
<!DOCTYPE html>
<html><head runat="server"><title></title>
<asp:ContentPlaceHolder ID="head" runat="server">
</asp:ContentPlaceHolder>
</head><body>
<form id="form1" runat="server"><div>
<table class="auto-style1"><tr>
<td class="auto-style3">
<asp:Image ID="Image1" runat="server" Height="122px" Width="145px"
ImageUrl="~/images/logo.png"/>
</td>
<td class="auto-style4" style="font-size: 77px">CollectiGrade</td>
</tr><tr>
<td class="auto-style10"><br />
<asp:HyperLink ID="HyperLink1" runat="server"
NavigateUrl="~/Home.aspx">Home</asp:HyperLink>
<br />
<asp:HyperLink ID="HyperLink2" runat="server" NavigateUrl="~/About_Us.aspx">About
Us</asp:HyperLink>
</td>
<td class="auto-style9"></td>
</tr><tr>
<td class="auto-style5">
&nbsp;</td>
<td rowspan="2">
<asp:ContentPlaceHolder ID="ContentPlaceHolder1" runat="server">
</asp:ContentPlaceHolder>
</td></tr><tr>
<td class="auto-style7">&nbsp;</td></tr><tr>
<td class="auto-style2">&nbsp;</td>
<td colspan="2">&copy; Kolte & Kotian Solution Pvt Ltd </td>
</tr></table></div></form></body></html>
```

Master_Page.Master.cs

```
using System;
using System.Web.UI;
namespace Practical_11
{
```

Name: Anchal Singh

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```
public partial class Site1 :  
System.Web.UI.MasterPage  
{  
protected void Page_Load(object sender, EventArgs e)  
{}}}
```

Home.aspx

```
<%@ Page Title="" Language="C#" MasterPageFile="~/Master_Page.Master"  
AutoEventWireup="true" CodeBehind="Home.aspx.cs" Inherits="Practical_11.WebForm1"  
%>  
<asp:Content ID="Content1" ContentPlaceHolderID="head" runat="server">  
</asp:Content>  
<asp:Content ID="Content2" ContentPlaceHolderID="ContentPlaceHolder1" runat="server">  
<label> Anchal Singh 119</label>  
<div>  
<strong style="font-size: 28px">Welcome to </strong><span style="font-size:  
28px"><strong><span class="auto-style1">CollectiGrade<br />  
<br />  
</span></strong>This is the HOME page.</span>  
</div></asp:Content>
```

Home.aspx.cs

```
using System;  
using System.Web.UI;  
namespace Practical_11  
{  
public partial class WebForm1 :  
System.Web.UI.Page  
{  
protected void Page_Load(object sender, EventArgs e)  
{}}}
```

About Us.aspx

```
<%@ Page Title="" Language="C#" MasterPageFile="~/Master_Page.Master"  
AutoEventWireup="true" CodeBehind="About_Us.aspx.cs"  
Inherits="Practical_11.About_Us" %>  
<asp:Content ID="Content1" ContentPlaceHolderID="head" runat="server">  
<style type="text/css"> .auto-style11 { text-decoration: underline;  
}  
</style>  
</asp:Content>  
<asp:Content ID="Content2" ContentPlaceHolderID="ContentPlaceHolder1" runat="server">  
<p style="font-size: 28px">
```


Name: Anchal Singh

Roll No: 119

```
<strong>Welcome to the <span class="auto-style11">CollectiGrade</span></strong></p>
<p style="font-size: 28px">
This is the ABOUT US Page.</p>
<b id="docs-internal-guid-39703627-7fff-c077-8a9e-77581ecafdd4" style="font-
weight:normal;">
<p dir="ltr" style="line-height:1.295;margin-right: -19.5pt;margin-top:0pt;margin-
bottom:8pt;">
<span class="auto-style11" style="font-size: medium; font-family: Cambria,serif; color:
#000000; background-color: transparent; font-weight: 700; font-style: normal; font-variant:
normal; vertical-align: baseline; white-space: pre; white-space: pre-wrap;">MOTTO:
“CollectiGrade: Where Antiques Find Their True Worth”</span></p></b><p style="font-size:
28px"> &nbsp;</p>
</asp:Content>
```

Output:



Name: Anchal Singh

Roll No: 119

Module 3: Database Programming ASP.NET

1. Create a web application to connect to a database using ADO.NET and display data in a GridView.(connected and disconnected)

Anchal119.aspx

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="Anchal119.aspx.cs"
Inherits="Anchal119.Anchal119" %> <!DOCTYPE html>
<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
<title>ADO.NET Practical</title>
</head><body>
<form id="form1" runat="server">
<h2>Anchal Singh 119</h2>
<asp:GridView ID="GridView1" runat="server" AutoGenerateColumns="False"
DataKeyNames="Id">
<Columns>
<asp:BoundField DataField="Id" HeaderText="ID" />
<asp:BoundField DataField="name" HeaderText="Name" />
<asp:BoundField DataField="age" HeaderText="Age" />
</Columns>
</asp:GridView>
<br />
<asp:Button ID="Connected" runat="server" Text="Connected" OnClick="Connected_Click"
/>
<asp:Button ID="Disconnected" runat="server" Text="Disconnected"
OnClick="Disconnected_Click" />
</form></body></html>
```

Anchal119.aspx.cs

```
using System;
using System.Data;
using System.Data.SqlClient;
using System.Web.UI;
namespace Anchal119
{
public partial class Anchal119 : System.Web.UI.Page
{
String StrSQL = "SELECT * FROM data";
String StrConnection = @"Data Source=DESKTOP-84HFCTF\SQLEXPRESS;Initial
Catalog=AnchalDB;Integrated Security=True";

protected void Page_Load(object sender, EventArgs e)
{
```

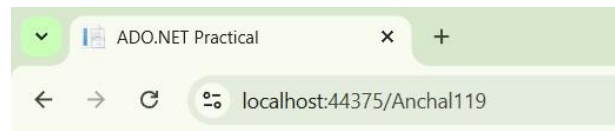
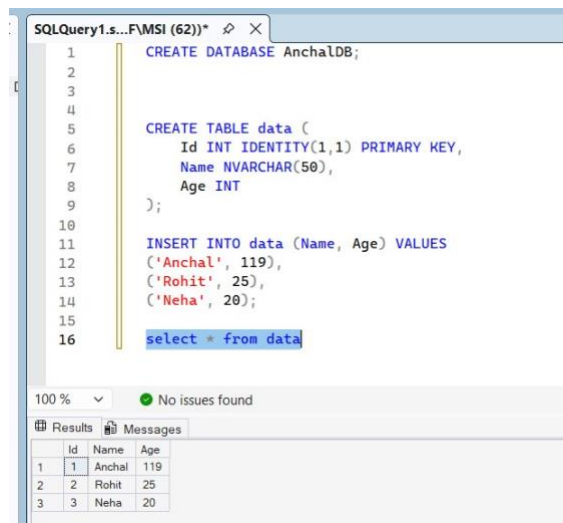
Name: Anchal Singh

Roll No: 119

```
// Nothing needed here unless you want auto load  
}
```

```
protected void Connected_Click(object sender, EventArgs e)  
{  
    using (SqlConnection objConn = new SqlConnection(StrConnection))  
    {  
        SqlCommand objCmd = new SqlCommand(StrSQL, objConn);           objConn.Open();  
        SqlDataReader objDr = objCmd.ExecuteReader();  
        GridView1.DataSource = objDr;  
        GridView1.DataBind();  
    } // Connection auto-closed with 'using'  
}  
  
protected void Disconnected_Click(object sender, EventArgs e)  
{  
    DataSet objDs = new DataSet();           using (SqlConnection objConn = new  
        SqlConnection(StrConnection))  
    {  
        SqlCommand objCmd = new SqlCommand(StrSQL, objConn);           SqlDataAdapter  
        objDa = new SqlDataAdapter(objCmd);           objDa.Fill(objDs, "data");  
    }  
    GridView1.DataSource = objDs.Tables[0];  
    GridView1.DataBind();  
}}}
```

Output:



Anchal Singh 119

ID	Name	Age
1	Anchal	119
2	Rohit	25
3	Neha	20

Connected

Disconnected

Name: Anchal Singh

Roll No: 119

2. Implement a form to insert, update, delete, and display student records using connected architecture.

```
create database AnchalDB;
CREATE TABLE student
(
Id INT IDENTITY(1,1) PRIMARY KEY, name NVARCHAR(50), course NVARCHAR(50),
marks INT
)
```

Student.aspx

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="Student.aspx.cs"
Inherits="Anchal119.Student" %> <!DOCTYPE html>
<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
<title>Student Record Management</title>
<style> body{ font-family: Arial; margin:40px;
} table{ border-collapse:collapse;
} td{ padding:8px;
}
</style> </head>
<body>
<form id="form1" runat="server">
<h3>Anchal 119</h3>
<h2>Student Record Management</h2>
<table><tr>
<td>Student ID</td>
<td>
<asp:TextBox ID="txtId" runat="server"></asp:TextBox>
</td></tr><tr>
<td>Name</td>
<td>
<asp:TextBox ID="txtName" runat="server"></asp:TextBox>
</td></tr><tr>
<td>Course</td>
<td>
<asp:TextBox ID="txtCourse" runat="server"></asp:TextBox>
</td></tr><tr>
<td>Marks</td><td>
<asp:TextBox ID="txtMarks" runat="server"></asp:TextBox>
</td></tr><tr>
<td colspan="2">
<asp:Button ID="btnInsert" runat="server" Text="Insert"
```

Name: Anchal Singh

Roll No: 119

```
OnClick="btnInsert_Click" />
<asp:Button ID="btnUpdate" runat="server" Text="Update"
OnClick="btnUpdate_Click" />
<asp:Button ID="btnDelete" runat="server" Text="Delete"
OnClick="btnDelete_Click" />
<asp:Button ID="btnDisplay" runat="server" Text="Display"
OnClick="btnDisplay_Click" />
<asp:Button ID="btnClear" runat="server" Text="Clear"
OnClick="btnClear_Click" />
</td></tr></table>
<br /><br />
<asp:GridView ID="GridView1" runat="server"
AutoGenerateColumns="True"
BorderWidth="1"
BorderColor="Black">
</asp:GridView>
</form> </body>
</html>
```

Student.aspx.cs

```
using System; using System.Data; using System.Data.SqlClient; using
System.Web.UI.WebControls; using System.Xml.Linq; namespace Anchal119
{
    public partial class Student : System.Web.UI.Page
    {
        SqlConnection con = new SqlConnection(
@"Data Source=DESKTOP-84HFCTF\SQLEXPRESS;Initial Catalog=AnchalDB;Integrated
Security=True");
        protected void Page_Load(object sender, EventArgs e)
        {
            if (!IsPostBack)
            {
                LoadData();
            }
        }
        void LoadData()
        {
            con.Open();
            SqlCommand cmd = new SqlCommand("select * from student", con); SqlDataReader dr =
            cmd.ExecuteReader();
            GridView1.DataSource = dr;
            GridView1.DataBind();
            con.Close();
        }
        void ClearData()
        {
            txtId.Text = "";
            txtName.Text = "";
        }
    }
}
```

Name: Anchal Singh

Roll No: 119

```
txtCourse.Text = "";
txtMarks.Text = "";
}
protected void btnInsert_Click(object sender, EventArgs e)
{
    con.Open();
    SqlCommand cmd = new SqlCommand(
        "insert into student values(@id,@name,@course,@marks)", con);
    cmd.Parameters.AddWithValue("@id", txtId.Text);
    cmd.Parameters.AddWithValue("@name", txtName.Text);
    cmd.Parameters.AddWithValue("@course", txtCourse.Text);
    cmd.Parameters.AddWithValue("@marks", txtMarks.Text);
    cmd.ExecuteNonQuery();
    con.Close();
    LoadData();
    ClearData();
}
protected void btnUpdate_Click(object sender, EventArgs e)
{
    con.Open();
    SqlCommand cmd = new SqlCommand(
        "update student set name=@name,course=@course,marks=@marks where id=@id", con);
    cmd.Parameters.AddWithValue("@id", txtId.Text);
    cmd.Parameters.AddWithValue("@name", txtName.Text);
    cmd.Parameters.AddWithValue("@course", txtCourse.Text);
    cmd.Parameters.AddWithValue("@marks", txtMarks.Text);
    cmd.ExecuteNonQuery();
    con.Close();
    LoadData();
    ClearData();
}
protected void btnDelete_Click(object sender, EventArgs e)
{
    con.Open();
    SqlCommand cmd = new SqlCommand( "delete from student where id=@id", con);
    cmd.Parameters.AddWithValue("@id", txtId.Text);
    cmd.ExecuteNonQuery();
    con.Close();
    LoadData();
    ClearData();
}
protected void btnDisplay_Click(object sender, EventArgs e)
{
    LoadData();
}
protected void btnClear_Click(object sender, EventArgs e)
```

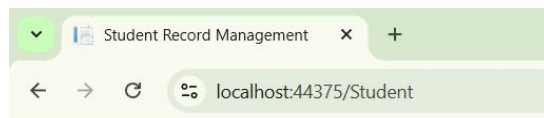
Name: Anchal Singh

Roll No: 119

```
{  
ClearData();  
}}}
```

Output:

Insert records



Anchal 119

Student Record Management

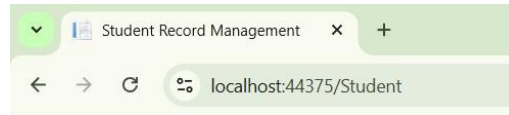
Student ID

Name

Course

Marks

Id	name	course	marks
119	Anchal Singh	FYMCA	80



Anchal 119

Student Record Management

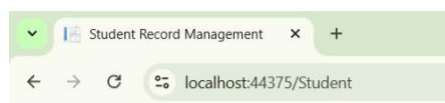
Student ID

Name

Course

Marks

Id	name	course	marks
119	Anchal Singh	FYMCA	80



Anchal 119

Student Record Management

Student ID

Name

Course

Marks

Id	name	course	marks
118	Rhea Lobo		0
119	Anchal Singh	FYMCA	80



Anchal 119

Student Record Management

Student ID

Name

Course

Marks

Id	name	course	marks
118	Rhea Lobo	SYMCA	90
119	Anchal Singh	FYMCA	80

Name: Anchal Singh

Roll No: 119

3. Design a web application using disconnected architecture (DataSet and DataAdapter) for managing data.

StudentDS.aspx

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="StudentDS.aspx.cs"
Inherits="Anchal119.StudentDS" %> <!DOCTYPE html>
<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
<title>Student Data Management (Disconnected)</title>
</head>
<body>
<form id="form1" runat="server">
<h2>Student Management - Disconnected Architecture</h2> Student ID :
<asp:TextBox ID="txtId" runat="server"></asp:TextBox> <br /><br /> Name :
<asp:TextBox ID="txtName" runat="server"></asp:TextBox> <br /><br /> Course :
<asp:TextBox ID="txtCourse" runat="server"></asp:TextBox> <br /><br /> Marks :
<asp:TextBox ID="txtMarks" runat="server"></asp:TextBox>
<br /><br />
<asp:Button ID="btnInsert" runat="server" Text="Insert" OnClick="btnInsert_Click" />
<asp:Button ID="btnUpdate" runat="server" Text="Update" OnClick="btnUpdate_Click" />
<asp:Button ID="btnDelete" runat="server" Text="Delete" OnClick="btnDelete_Click" />
<asp:Button ID="btnDisplay" runat="server" Text="Display" OnClick="btnDisplay_Click" />
<br /><br />
<asp:GridView ID="GridView1" runat="server"></asp:GridView>
</form>
</body>
</html>
```

StudentDS.aspx.cs

using System; using System.Data; using System.Data.SqlClient; using System.Xml.Linq;
namespace Anchal119

```
{    public partial class StudentDS : System.Web.UI.Page
{
    SqlConnection con = new SqlConnection(
```

```
@ "Data Source=DESKTOP-84HFCTF\SQLEXPRESS;Initial Catalog=AnchalDB;Integrated
Security=True");
```

```
    SqlDataAdapter da;
```

```
    DataSet ds;
```

```
    protected void btnDisplay_Click(object sender, EventArgs e)
```

```
{
```

```
    da = new SqlDataAdapter("select * from student", con);
```

```
    ds = new DataSet();
```


Name: Anchal Singh

Roll No: 119

```
da.Fill(ds, "student");
GridView1.DataSource = ds.Tables["student"];
GridView1.DataBind();
}
protected void btnInsert_Click(object sender, EventArgs e)
{
da = new SqlDataAdapter("select * from student", con);
SqlCommandBuilder cb = new SqlCommandBuilder(da);
ds = new DataSet();
da.Fill(ds, "student");
DataRow dr = ds.Tables["student"].NewRow();
dr["Id"] = txtId.Text;
dr["name"] = txtName.Text;
dr["course"] = txtCourse.Text;
dr["marks"] = txtMarks.Text;
ds.Tables["student"].Rows.Add(dr);
da.Update(ds, "student");
GridView1.DataSource = ds.Tables["student"];
GridView1.DataBind();
}
protected void btnUpdate_Click(object sender, EventArgs e)
{
da = new SqlDataAdapter("select * from student", con);
SqlCommandBuilder cb = new SqlCommandBuilder(da);
ds = new DataSet();
da.Fill(ds, "student");
foreach (DataRow dr in ds.Tables["student"].Rows)
{
if (dr["Id"].ToString() == txtId.Text)
{
dr["name"] = txtName.Text;
dr["course"] = txtCourse.Text;
dr["marks"] = txtMarks.Text;
}
}
da.Update(ds, "student");
GridView1.DataSource = ds.Tables["student"];
GridView1.DataBind();
}
protected void btnDelete_Click(object sender, EventArgs e)
{
da = new SqlDataAdapter("select * from student", con);
SqlCommandBuilder cb = new SqlCommandBuilder(da); ds = new DataSet();
da.Fill(ds, "student");          foreach (DataRow dr in ds.Tables["student"].Rows)

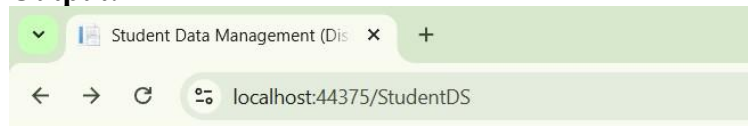
{
```

Name: Anchal Singh

Roll No: 119

```
if (dr["Id"].ToString() == txtId.Text)
{
    dr.Delete();
}
da.Update(ds, "student");
GridView1.DataSource = ds.Tables["student"];
GridView1.DataBind();
}
}
}
```

Output:



Student Management - Disconnected Architecture

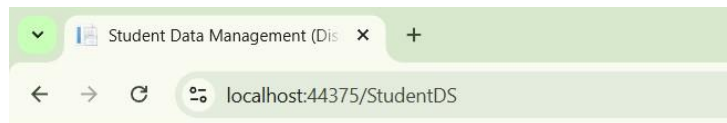
Student ID :

Name :

Course :

Marks :

Id	name	course	marks
119	Anchal Singh	FYMCA	80
1	Anchal Singh	FYMCA	80



Student Management - Disconnected Architecture

Student ID :

Name :

Course :

Marks :

Id	name	course	marks
1	Anchal	SYMCA	90
119	Anchal Singh	FYMCA	80

Name: Anchal Singh

Roll No: 119

4. Develop a stored procedure to fetch employee details and display them using DetailsView control.

```
/*Create Employee Table */  
CREATE TABLE employee  
(  
Id INT PRIMARY KEY, name VARCHAR(50), department VARCHAR(50), salary INT  
);
```

```
/*Insert sample data*/  
INSERT INTO employee VALUES (1,'Anchal','IT',50000);  
INSERT INTO employee VALUES (2,'Rhea','HR',45000);  
INSERT INTO employee VALUES (3,'Alia','Finance',60000);
```

```
/*Create Stored Procedure */
```

```
CREATE PROCEDURE GetEmployee  
@EmpId INT  
AS  
BEGIN  
SELECT * FROM employee WHERE Id = @EmpId  
END
```

Employee.aspx

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="Employee.aspx.cs"  
Inherits="Anchal119.Employee" %> <!DOCTYPE html>  
<html xmlns="http://www.w3.org/1999/xhtml">  
<head runat="server">  
<title>Employee Details</title>  
</head>  
<body>  
<form id="form1" runat="server"> <h2>Employee Details - 119</h2> Enter Employee Id :  
<asp:TextBox ID="txtId" runat="server"></asp:TextBox>  
<asp:Button ID="btnSearch" runat="server"  
Text="Search"  
OnClick="btnSearch_Click" />  
<br /><br />  
<asp:DetailsView ID="DetailsView1" runat="server" AutoGenerateRows="True"  
Width="300px">  
</asp:DetailsView></form>  
</body></html>
```

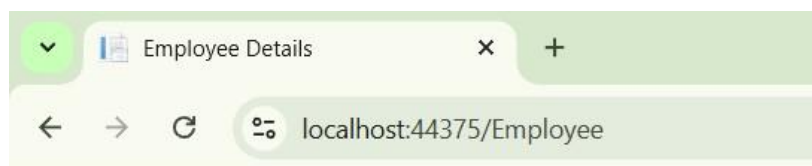
Name: Anchal Singh

Roll No: 119

Employee.aspx.cs

```
using System;
using System.Data;
using System.Data.SqlClient;
using System.Web.UI.WebControls;
using System.Data;
using System.Data.SqlClient;
namespace Anchal119
{
    public partial class Employee : System.Web.UI.Page
    {
        string conStr = @"Data Source=DESKTOP-84HFCTF\SQLEXPRESS;Initial
        Catalog=AnchalDB;Integrated Security=True";
        protected void btnSearch_Click(object sender, EventArgs e)
        {
            SqlConnection con = new SqlConnection(conStr);
            SqlCommand cmd = new SqlCommand("GetEmployee", con);
            cmd.CommandType = CommandType.StoredProcedure;
            cmd.Parameters.AddWithValue("@EmpId", txtId.Text);
            SqlDataAdapter da = new SqlDataAdapter(cmd);
            DataTable dt = new DataTable();
            da.Fill(dt);
            DetailsView1.DataSource = dt;
            DetailsView1.DataBind();
        }
    }
}
```

Output:



Employee Details - 119

Enter Employee Id :

Id	1
name	Anchal
department	IT
salary	50000

Name: Anchal Singh

Roll No: 119

5. Web Application to demonstrate LINQ with object data source.

Student.cs

```
using System;
using System.Collections.Generic;
using System.Linq;
namespace Anchal119
{
    public class StudentModel
    {
        public int Id { get; set; }
        public string Name { get; set; }
        public int Marks { get; set; }
    }
    public class StudentData
    {
        public List<StudentModel> GetStudents()
        {
            List<StudentModel> students = new List<StudentModel>()
            {
                new StudentModel { Id = 1, Name = "Anchal", Marks = 85 },
                new StudentModel { Id = 2, Name = "Aman", Marks = 78 },
                new StudentModel { Id = 3, Name = "Rahul", Marks = 92 },
                new StudentModel { Id = 4, Name = "Neha", Marks = 79 }
            };
            var result = from s in students where s.Marks > 80 select s;
            return result.ToList();
        }
    }
}
```

LINQDemo.aspx

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="LINQDemo.aspx.cs"
Inherits="Anchal119.LINQDemo"
```

```
%>
```

```
<!DOCTYPE html>
<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
<title>LINQ with ObjectDataSource</title>
</head>
<body>
<form id="form1" runat="server">
<h2>LINQ with ObjectDataSource Example</h2>
<asp:GridView ID="GridView1"
```

Name: Anchal Singh

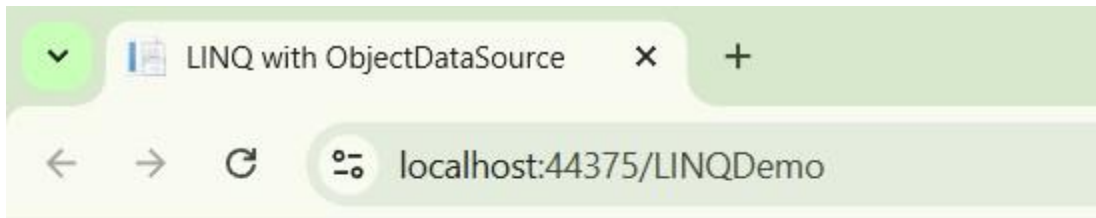
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```
runat="server"
AutoGenerateColumns="True"
DataSourceID="ObjectDataSource1">
</asp:GridView>
<asp:ObjectDataSource ID="ObjectDataSource1" runat="server"
TypeName="Anchal119.StudentData" SelectMethod="GetStudents">
</asp:ObjectDataSource>
</form></body></html>
```

LINQDemo.aspx.cs

```
using System;
namespace Anchal119
{
    public partial class LINQDemo :
        System.Web.UI.Page
    {
        protected void Page_Load(object sender, EventArgs e)
        {}
    }
}
```

Output:



LINQ with ObjectDataSource Example

Id	Name	Marks
1	Anchal	85
3	Rahul	92

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6. Web Application to demonstrate LINQ with a data set.

LINQDataSet.aspx

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="LINQDataSet.aspx.cs"
Inherits="Anchal119.LINQDataSet" %>
<!DOCTYPE html>
<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
<title>LINQ with DataSet</title>
</head>
<body>
<form id="form1" runat="server">
<h2>LINQ with DataSet Example</h2>
<asp:Button ID="btnShow" runat="server" Text="Show Students" OnClick="btnShow_Click"
/> <br /><br />
<asp:GridView ID="GridView1" runat="server"></asp:GridView>
</form>
</body>
</html>
```

LINQDataSet.aspx.cs

```
using System;
using System.Data;
using System.Data.SqlClient;
using System.Linq;
using System.Xml.Linq;
namespace Anchal119
{
public partial class LINQDataSet : System.Web.UI.Page
{
SqlConnection con = new SqlConnection(
@"Data Source=DESKTOP-84HFCTF\SQLEXPRESS;Initial Catalog=AnchalDB;Integrated
Security=True");    protected void btnShow_Click(object sender, EventArgs e)
{
SqlDataAdapter da = new SqlDataAdapter("select * from student", con);
DataSet ds = new DataSet();
da.Fill(ds, "student");
var result = from s in ds.Tables["student"].AsEnumerable() where s.Field<int>("marks") > 80

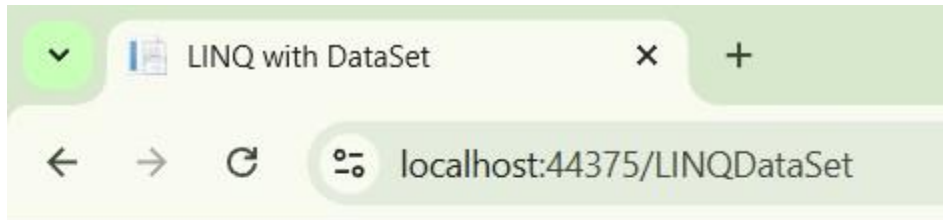
select s;
GridView1.DataSource = result.CopyToDataTable();
GridView1.DataBind();
}
```

Name: Anchal Singh

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```
}  
}
```

Output:



LINQ with DataSet Example

Show Students

Id	name	course	marks
1	Anchal	SYMCA	90

Name: Anchal Singh

Roll No: 119

7. Create a webpage that demonstrates the use of data bound controls

DataBound.aspx

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="DataBound.aspx.cs"
Inherits="Anchal119.DataBound" %> <!DOCTYPE html>
<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
<title>Data Bound</title>
</head>
<body>
<form id="form1" runat="server">
<p>Anchal 119</p>
<h2>Student Details using Data Bound</h2>
<asp:Button ID="btnLoad" runat="server"
Text="Load Data"
OnClick="btnLoad_Click" />
<br /><br />
<h3>GridView</h3>
<asp:GridView ID="GridView1" runat="server"></asp:GridView>
<br />
<h3>DetailsView</h3>
<asp:DetailsView ID="DetailsView1" runat="server"
AutoGenerateRows="True">
</asp:DetailsView>
<br />
<h3>DropDownList</h3>
<asp:DropDownList ID="DropDownList1" runat="server"></asp:DropDownList> </form>
</body>
</html>
```

DataBound.aspx.cs

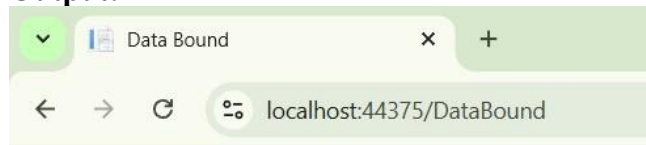
```
using System;
using System.Data;
using System.Data.SqlClient;
namespace Anchall19
{
    public partial class DataBound : System.Web.UI.Page
    {
        SqlConnection con = new SqlConnection(
@"Data Source=DESKTOP-84HFCTF\SQLEXPRESS;Initial Catalog=AnchalDB;Integrated
Security=True");
        protected void btnLoad_Click(object sender, EventArgs e)
```

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```
{  
SqlDataAdapter da = new SqlDataAdapter("select * from student", con); DataSet ds = new  
DataSet();      da.Fill(ds, "student");  
GridView1.DataSource = ds.Tables["student"];  
GridView1.DataBind();  
DetailsView1.DataSource = ds.Tables["student"];  
DetailsView1.DataBind();  
DropDownList1.DataSource = ds.Tables["student"];  
DropDownList1.DataTextField = "name";  
DropDownList1.DataValueField = "Id";  
DropDownList1.DataBind();  
}  
}  
}
```

Output:



Anchal 119

Student Details using Data Bound

Load Data

GridView

Id	name	course	marks
1	Anchal	SYMCA	90
119	Anchal Singh	FYMCA	80

DetailsView

Id	1
name	Anchal
course	SYMCA
marks	90

DropDownList

Anchal Singh ▼

Name: Anchal Singh

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9. Design a web page to perform CRUD operation using SqlDataSource

create database AnchalDB

```
CREATE TABLE student
(
  Id INT IDENTITY(1,1) PRIMARY KEY,
  Name NVARCHAR(50),
  Course NVARCHAR(50),
  Marks INT
)
```

Crud.aspx

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="CRUD.aspx.cs"
Inherits="Anchal.CRUD" %>
<!DOCTYPE html>
<html>
<head runat="server">
<title>Student CRUD using SqlDataSource</title>
<style>
body{ font-family: Arial; margin:40px; }
h2{ color:#2c3e50;
} .grid{ margin-top:20px; }.insertbox{ margin-top:30px; padding:10px; border:1px solid #ccc;
width:300px; }
</style>
</head>
<body>
<form id="form1" runat="server">
<h2>Student CRUD using SqlDataSource</h2>
<!-- GRIDVIEW FOR DISPLAY / EDIT / DELETE -->
<asp:GridView ID="GridView1" runat="server" CssClass="grid"
DataSourceID="SqlDataSource1" AutoGenerateColumns="False"
AutoGenerateEditButton="True" AutoGenerateDeleteButton="True" DataKeyNames="Id">
<Columns> <asp:BoundField DataField="Id" HeaderText="Id" ReadOnly="True"/>
<asp:BoundField DataField="Name" HeaderText="Name"/>
<asp:BoundField DataField="Course" HeaderText="Course"/>
<asp:BoundField DataField="Marks" HeaderText="Marks"/>
</Columns> </asp:GridView>
<br /><br />
<!-- DETAILS VIEW FOR INSERT -->
<asp:DetailsView ID="DetailsView1" runat="server" CssClass="insertbox"
DataSourceID="SqlDataSource1"
AutoGenerateRows="True" DefaultMode="Insert" AutoGenerateInsertButton="True">
```

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```
</asp:DetailsView>
<!-- SQLDATASOURCE -->
<asp:SqlDataSource ID="SqlDataSource1" runat="server"
ConnectionString="<%$ ConnectionStrings:AnchalDBConnectionString %>"
SelectCommand="SELECT * FROM student"
InsertCommand="INSERT INTO student(Name,Course,Marks)
VALUES(@Name,@Course,@Marks)"
UpdateCommand="UPDATE student SET Name=@Name,Course=@Course,Marks=@Marks
WHERE Id=@Id"
DeleteCommand="DELETE FROM student WHERE Id=@Id"> </asp:SqlDataSource>

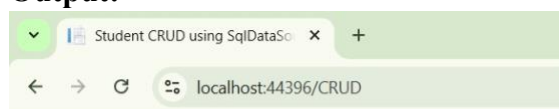
</form> </body> </html>
```

Crud.aspx.cs

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;

namespace Anchal119
{
    public partial class CRUD :
        System.Web.UI.Page
    {
        protected void Page_Load(object sender, EventArgs e)
        {}
    }
}
```

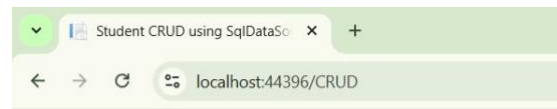
Output:



Student CRUD using SqlDataSource

	Id	Name	Course	Marks
Edit Delete	1	Anchal Singh	MCA	95

Id	2
Name	Rhea
Course	SYMCA
Marks	80
Insert Cancel	



Student CRUD using SqlDataSource

	Id	Name	Course	Marks
Edit Delete	1	Anchal Singh	MCA	95
Edit Delete	3	Rhea	SYMCA	80

Id	
Name	
Course	
Marks	
Insert Cancel	

Name: Anchal Singh

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Module 4: State Management and AJAX

1. Create a web application demonstrating client-side state management using ViewState and Cookies.

StateManagement.aspx

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="StateManagement.aspx.cs"
Inherits="StateManagementExample.StateManagement" %>
```

```
<!DOCTYPE html>
<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
<title>Client-Side State Management</title>
<style>    label {
font-weight: bold;
}
</style>
</head>
<body>
<form id="form1" runat="server">
<div>
<h1>Client-Side State Management with ViewState and Cookies</h1>

<!-- ViewState Section -->
<h2>Store Name using ViewState</h2>
<label for="txtNameViewState">Enter your name:</label>
<asp:TextBox ID="txtNameViewState" runat="server"></asp:TextBox>
<asp:Button ID="btnSaveViewState" runat="server" Text="Save to ViewState"
OnClick="btnSaveViewState_Click" />
<br /><br />
<asp:Label ID="lblViewState" runat="server" Text="Your name from ViewState will appear
here."></asp:Label>
<br /><br />

<!-- Cookies Section -->
<h2>Store Name using Cookies</h2>
<label for="txtNameCookie">Enter your name:</label>
<asp:TextBox ID="txtNameCookie" runat="server"></asp:TextBox>
<asp:Button ID="btnSaveCookie" runat="server" Text="Save to Cookie"
OnClick="btnSaveCookie_Click" />
<br /><br />
```

Name: Anchal Singh

Roll No: 119

```
<asp:Label ID="lblCookie" runat="server" Text="Your name from Cookies will appear here."></asp:Label>
```

```
</div>
```

```
</form>
```

```
</body>
```

```
</html>
```

StateManagement.aspx.cs

```
using System;
```

```
using System.Web;
```

```
namespace StateManagementExample
```

```
{
```

```
public partial class StateManagement : System.Web.UI.Page
```

```
{
```

```
protected void Page_Load(object sender, EventArgs e)
```

```
{
```

```
// Check if ViewState contains a stored name and display it
```

```
if (ViewState["NameFromViewState"] != null)
```

```
{
```

```
lblViewState.Text = "Your name from ViewState: " +
```

```
ViewState["NameFromViewState"].ToString();
```

```
}
```

```
// Check if Cookies contains a stored name and display it
```

```
if (Request.Cookies["NameFromCookie"] != null)
```

```
{
```

```
lblCookie.Text = "Your name from Cookie: " +
```

```
Request.Cookies["NameFromCookie"].Value;
```

```
}
```

```
}
```

```
// Save name to ViewState
```

```
protected void btnSaveViewState_Click(object sender, EventArgs e)
```

```
{
```

```
string nameFromViewState = txtNameViewState.Text;
```

```
ViewState["NameFromViewState"] = nameFromViewState;
```

```
lblViewState.Text = "Your name has been saved to ViewState: " + nameFromViewState;
```

```
}
```

```
// Save name to Cookie
```

```
protected void btnSaveCookie_Click(object sender, EventArgs e)
```

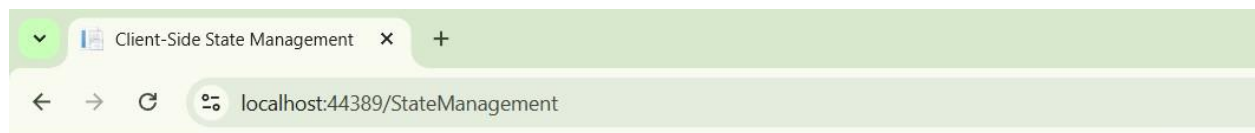
```
{
```

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```
string nameFromCookie = txtNameCookie.Text;
HttpCookie cookie = new HttpCookie("NameFromCookie", nameFromCookie)
{
    Expires = DateTime.Now.AddDays(1) // Cookie will expire in 1 day
};
Response.Cookies.Add(cookie);
lblCookie.Text = "Your name has been saved to Cookie: " + nameFromCookie;
}
}
}
```

Output:



Client-Side State Management with ViewState and Cookies

Store Name using ViewState

Enter your name:

Your name from ViewState: Anchal

Store Name using Cookies

Enter your name:

Your name has been saved to Cookie: Anchal Singh

Name: Anchal Singh

Roll No: 119

2. Design a session-based login system where users are redirected to a dashboard upon successful login.

File → New → Project->ASP.NET Web Application (.NET Framework)->MVC template

Models folder → Add → Class → name

User.cs public class User

```
{    public string Username { get; set;
}    public string Password { get; set;
}
}
```

Controllers → Add → Controller → MVC 5 Controller → Empty ->AccountContoller

```
using System.Linq;
```

```
using System.Collections.Generic; using
System.Web.Mvc;
```

```
public class AccountController : Controller
```

```
{
```

```
private List<User> users = new List<User>
```

```
{
```

```
new User { Username = "anchalsingh", Password = "password123" }
```

```
};
```

```
public ActionResult Login()
```

```
{
```

```
return View(new User());
```

```
}
```

```
[HttpPost]
```

```
public ActionResult Login(User model)
```

```
{
```

```
var user = users.FirstOrDefault(u => u.Username == model.Username && u.Password ==
model.Password);
```

```
if (user != null)
```

```
{
```

```
Session["Username"] = model.Username;          return
```

```
RedirectToAction("Dashboard");
```

```
}
```


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```
ViewBag.ErrorMessage = "Invalid username or password.";    return
View(model);
}

public ActionResult Dashboard()
{
    if (Session["Username"] == null)
        return RedirectToAction("Login");
    ViewBag.Username = Session["Username"].ToString();
    return View();
}

public ActionResult Logout()
{
    Session.Clear();
    return RedirectToAction("Login");
}}
```

Login() method → Add View

View Name: Login Template: Empty Model class: User

@model User

<h2>Login</h2>

@if(ViewBag.ErrorMessage != null)

{

<p style="color:red">@ViewBag.ErrorMessage</p>

}

@using (Html.BeginForm())

{<div>

<label>Username:</label>

@Html.TextBoxFor(m => m.Username)

</div><div>

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```
<label>Password:</label>
```

```
@Html.PasswordFor(m => m.Password) </div>
```

```
<button type="submit">Login</button>
```

```
}
```

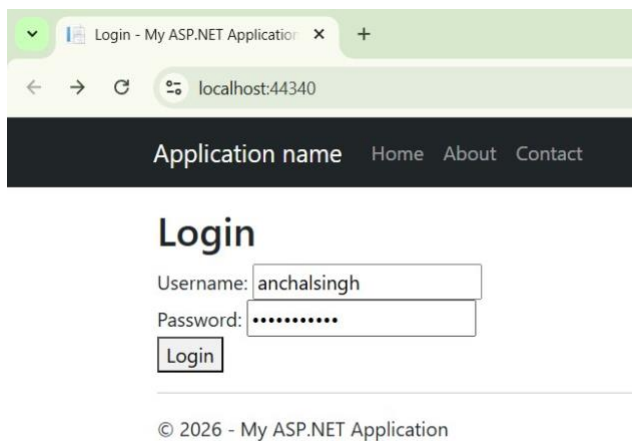
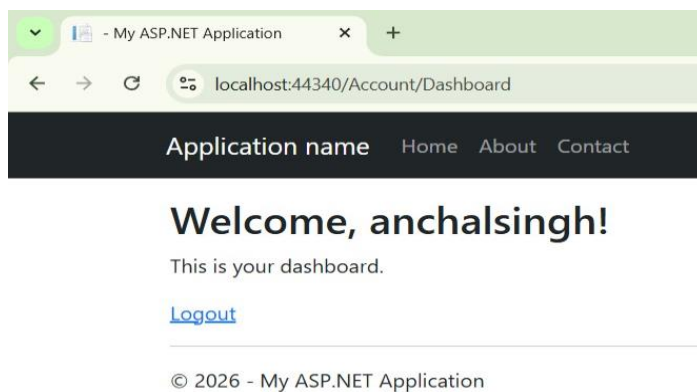
Dashboard() method → Add View → Empty View

```
<h2>Welcome, @ViewBag.Username!</h2>
```

```
<p>This is your dashboard.</p>
```

```
@Html.ActionLink("Logout", "Logout", "Account") App_Start / RouteConfig.cs defaults: new {  
controller = "Account", action = "Login", id = UrlParameter.Optional }
```

Output:



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3. Implement an AJAX-enabled web form to fetch user details without reloading the page.

Default.aspx

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="Default.aspx.cs"
Inherits="AJAXExample._Default" %>

<!DOCTYPE html>
<html>
<head runat="server">
<title>AJAX Fetch User Details - Anchal Singh 119</title>
<script src="https://ajax.googleapis.com/ajax/libs/jquery/3.5.1/jquery.min.js"></script> <script
type="text/javascript">
<h1>AJAX Fetch User Details - Anchal Singh 119</h1>
function fetchUserDetails() {
var userID = $("#userID").val();

if (!userID) {
alert("Please enter a User ID");
return;
}

$("#userDetails").html("Loading...");

$.ajax({ type: "POST", url:
"/Default.aspx/GetUserDetails",          data:
JSON.stringify({ userID: userID }),
contentType: "application/json; charset=utf-8",
dataType: "json",          success: function (response)
{
if (response.d) {          var user = response.d;
$("#userDetails").html(
"Name: " + user.Name + "<br/>" +          "Email:
" + user.Email + "<br/>" +
"Phone: " + user.Phone
);
} else {
$("#userDetails").html("User not found.");
}
},
error: function (error) {
console.log(error);
$("#userDetails").html("An error occurred. Check
console.");
}
```

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```
}
});
}
</script>
</head>
<body>
<form id="form1" runat="server">
<h1>AJAX Fetch User Details - Anchal Singh 119</h1>
<div>
<label for="userID">Enter User ID:</label>
<input type="text" id="userID" />
<button type="button" onclick="fetchUserDetails()">Fetch Details</button>    </div>

<div id="userDetails" style="margin-top:10px;">
<!-- User details will be displayed here -->
</div>
</form>
</body>
</html>
```

Default.aspx.cs

```
using System; using
System.Web.Services;
using System.Web.UI;
namespace AJAXExample
{
public partial class _Default : Page
{
// Simple User class
public class User
{
public string Name { get; set; }      public
string Email { get; set; }
public string Phone { get; set; }
}
// Static method to simulate database fetch
private static User GetUserDetailsById(int userID)
{
if (userID == 1)
{
return new User
{
Name = "John Doe",
```

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```
Email = "john.doe@example.com",  
Phone = "123-456-7890"  
};  
}  
else if (userID == 2)  
{  
return new User  
{  
Name = "Jane Smith",  
Email = "jane.smith@example.com",  
Phone = "987-654-3210"  
};  
}  
return null; // User not found  
}
```

```
[WebMethod]  
public static object GetUserDetails(string userID)  
{  
if (!int.TryParse(userID, out int id))  
{  
return null; // Invalid input  
}  
  
return GetUserDetailsById(id);}}
```

Output:



AJAX Fetch User Details - Anchal Singh 119

Enter User ID:

Name: John Doe

Email: John.doe@example.com

Phone: 123-456-7890

Name: Anchal Singh

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4. Develop a simple to-do list application using AJAX controls and update panels.

Default.aspx:

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="Default.aspx.cs"
Inherits="StateManagementExample.Default" %>

<!DOCTYPE html>

<html xmlns="http://www.w3.org/1999/xhtml">

<head runat="server">

<title>To-Do List</title>

</head>

<body>

<form id="form1" runat="server">

<asp:ScriptManager ID="ScriptManager1" runat="server" />

<h2>To-Do List : Anchal Singh 119</h2>

<div>

<input type="text" id="taskInput" runat="server" />

<asp:Button ID="btnAddTask" runat="server" Text="Add Task" OnClick="btnAddTask_Click"
/>

</div>

<br />

<asp:UpdatePanel ID="UpdatePanel1" runat="server">

<ContentTemplate>
<asp:Repeater ID="rptTasks" runat="server">

<ItemTemplate>

<div style="background:#eee; padding:6px; margin-bottom:5px;">
```

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```
<%# Container.DataItem %>
```

```
<asp:Button ID="btnRemove" runat="server" Text="Remove"
```

```
CommandArgument="<%# Container.ItemIndex %>"
```

```
OnClick="RemoveTask_Click" />
```

```
</div>
```

```
</ItemTemplate>
```

```
</asp:Repeater>
```

```
</ContentTemplate>
```

```
</asp:UpdatePanel>
```

```
</form></body></html>
```

Default.aspx.cs:

```
using System;
```

```
using System.Collections.Generic;
```

```
namespace StateManagementExample
```

```
{
```

```
public partial class Default :
```

```
System.Web.UI.Page
```

```
{
```

```
private List<string> Tasks
```

```
{
```

```
get
```

```
{
```

```
if (Session["Tasks"] == null)
```

```
    Session["Tasks"] = new List<string>();
```

```
    return (List<string>)Session["Tasks"];
```

```
}
```

```
}
```

```
protected void Page_Load(object sender, EventArgs e)
```

```
{    if
```

```
(!IsPostBack)
```

Name: Anchal Singh

Roll No: 119

```
BindTasks();

}

protected void btnAddTask_Click(object sender, EventArgs e){

    string task = taskInput.Value.Trim();
    if (!string.IsNullOrEmpty(task))
    {

        Tasks.Add(task);
        taskInput.Value = "";
        BindTasks();}}

protected void RemoveTask_Click(object sender, EventArgs e){

    int index = int.Parse((sender as
    System.Web.UI.WebControls.Button).CommandArgument);

    Tasks.RemoveAt(index);

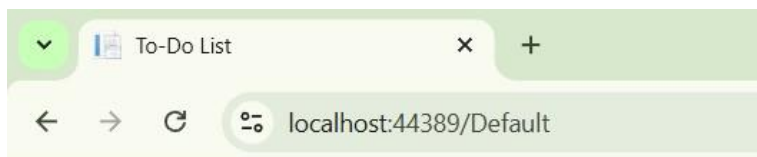
    BindTasks();}

private void BindTasks()

{

    rptTasks.DataSource = Tasks;          rptTasks.DataBind();
    }}}}
```

Output:



To-Do List : Anchal Singh 119

Go for walk

Study

Name: Anchal Singh

Roll No: 119

5. Demonstrate the use of the Global.asax file for handling application-level events.

Global.asax

```
<%@ Application Language="C#" CodeBehind="Global.asax.cs" Inherits="Module_4.Global"
%>
<!DOCTYPE html>
<html>
<head runat="server">
<title>Global.asax Example</title>
</head>
<body>
<h2>Global.asax Event Handling</h2>
</body> </html>
```

Global.asax.cs

This file contains the C# code for handling the application-level events.

```
using System;
using System.Web;
namespace Module_4
{
    public class Global : HttpApplication
    {
        // Runs when the application starts
        protected void Application_Start(object sender, EventArgs e)
        {
            System.Diagnostics.Debug.WriteLine("Application Started.");
        }

        // Runs when the application ends
        protected void Application_End(object sender, EventArgs e)
        {
            System.Diagnostics.Debug.WriteLine("Application Ended.");
        }

        // Runs when an unhandled error occurs
        protected void Application_Error(object sender, EventArgs e)
        {
            Exception exception = Server.GetLastError();
```

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```
System.Diagnostics.Debug.WriteLine("Application Error: " + exception.Message);
Server.ClearError(); // optional
}
// Runs when a new session starts
protected void Session_Start(object sender, EventArgs e)
{
    System.Diagnostics.Debug.WriteLine("Session Started.");
}
// Runs when a session ends
protected void Session_End(object sender, EventArgs e)
{
    System.Diagnostics.Debug.WriteLine("Session Ended.");}}} }
```

Output:

```
Application Started.
'iisexpress.exe' (CLR v4.0.30319: /LM/W3SVC/2/ROOT-1-134187125218503791): Loaded 'C:\WINDOWS\Micros
'iisexpress.exe' (CLR v4.0.30319: /LM/W3SVC/2/ROOT-1-134187125218503791): Loaded 'C:\WINDOWS\Micros
'iisexpress.exe' (CLR v4.0.30319: /LM/W3SVC/2/ROOT-1-134187125218503791): Loaded 'C:\WINDOWS\Micros
'iisexpress.exe' (CLR v4.0.30319: /LM/W3SVC/2/ROOT-1-134187125218503791): Loaded 'A_42a876a7_4be1_4
'iisexpress.exe' (CLR v4.0.30319: /LM/W3SVC/2/ROOT-1-134187125218503791): Loaded 'C:\WINDOWS\Micros
'iisexpress.exe' (CLR v4.0.30319: /LM/W3SVC/2/ROOT-1-134187125218503791): Loaded 'C:\WINDOWS\Micros
'iisexpress.exe' (CLR v4.0.30319: DefaultDomain): Loaded 'C:\WINDOWS\Microsoft.Net\assembly\GAC_MSI
'iisexpress.exe' (CLR v4.0.30319: /LM/W3SVC/2/ROOT-1-134187125218503791): Loaded 'C:\Users\MSI\AppData
'iisexpress.exe' (CLR v4.0.30319: /LM/W3SVC/2/ROOT-1-134187125218503791): Loaded 'C:\Users\MSI\AppData
Session Started.
'iisexpress.exe' (CLR v4.0.30319: /LM/W3SVC/2/ROOT-1-134187125218503791): Loaded 'C:\WINDOWS\Micros
Exception thrown: 'System.Web.HttpUnhandledException' in System.Web.dll
Application Error: Exception of type 'System.Web.HttpUnhandledException' was thrown.
The thread 9076 has exited with code 0 (0x0).
The thread 19212 has exited with code 0 (0x0).
The thread 14192 has exited with code 0 (0x0).
The thread 13560 has exited with code 0 (0x0).
The thread 7944 has exited with code 0 (0x0).
The thread 11828 has exited with code 0 (0x0).
The program '[7824] iisexpress.exe' has exited with code 4294967295 (0xffffffff).
```

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Module 5: Web Services and WCF

1. Create an XML-based web service to fetch product details and consume it in a web application.

ProductService.asmx(Web Service)

```
using System;
using System.Web.Services;
using System.Xml;
namespace SOAP_119
{
    [WebService(Namespace = "http://example.com/ProductService")]
    [WebServiceBinding(ConformsTo = WsiProfiles.BasicProfile1_1)]
    public class ProductService : WebService
    {
        [WebMethod]
        public XmlDocument GetProductDetails(int productId)
        {
            XmlDocument xmlDoc = new XmlDocument();
            XmlNode root = xmlDoc.CreateElement("ProductDetails");
            xmlDoc.AppendChild(root);
            XmlNode product = xmlDoc.CreateElement("Product");
            root.AppendChild(product);
            XmlNode id = xmlDoc.CreateElement("ProductId");
            id.InnerText = productId.ToString();
            product.AppendChild(id);
            XmlNode name = xmlDoc.CreateElement("ProductName");
            name.InnerText = "Product " + productId;
            product.AppendChild(name);
            XmlNode price = xmlDoc.CreateElement("Price");
            price.InnerText = (productId * 10).ToString();
            product.AppendChild(price);
            return xmlDoc;
        }
    }
}
```

ProductDetails.aspx

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="ProductDetails.aspx.cs"
Inherits="Soap_client_119.ProductDetails" %>
<!DOCTYPE html>
<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
<title>Product Details</title>
</head>
<body>
```

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```
<form id="form1" runat="server">
<div>
<h2>Product Details</h2>
<asp:Label ID="lblProductDetails" runat="server"
Text="Enter Product ID:" />
<asp:TextBox ID="txtProductId" runat="server"></asp:TextBox>

<asp:Button ID="btnFetchProduct" runat="server"
Text="Fetch Product"
OnClick="btnFetchProduct_Click" /><br /><br />
<asp:GridView ID="gvProductDetails" runat="server"></asp:GridView>    </div>
</form>
</body>
</html>
```

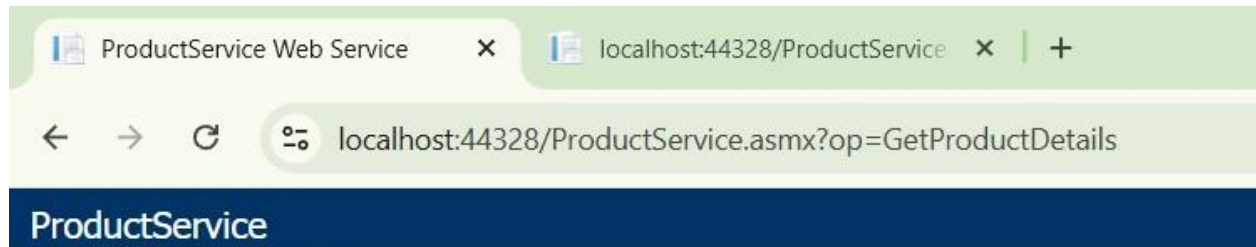
ProductDetails.aspx.cs

```
using System; using System.Web.UI; using System.IO; using System.Data;
using Soap_client_119.ServiceReference1; namespace Soap_client_119
{
public partial class ProductDetails : Page
{
protected void btnFetchProduct_Click(object sender, EventArgs e)
{
int productId = Convert.ToInt32(txtProductId.Text);
ProductServiceSoapClient service = new ProductServiceSoapClient(); var productXml =
service.GetProductDetails(productId);
DataSet ds = new DataSet();
StringReader sr = new StringReader(productXml.ToString()); ds.ReadXml(sr);
gvProductDetails.DataSource = ds.Tables[0];          gvProductDetails.DataBind();
}
}
}
```

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Output:



Click [here](#) for a complete list of operations.

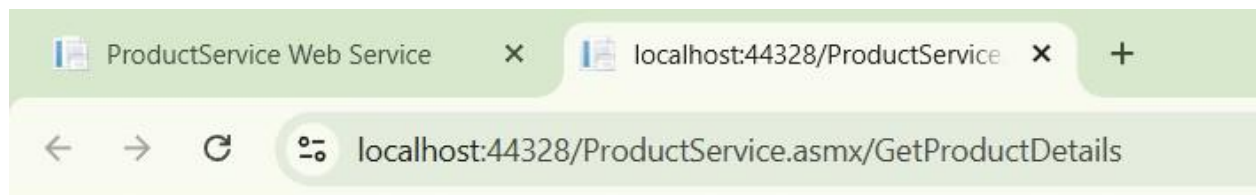
GetProductDetails

Test

To test the operation using the HTTP POST protocol, click the 'Invoke' button.

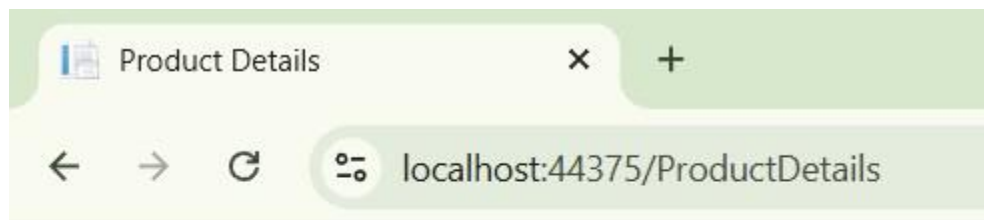
Parameter	Value
productId:	1

SOAP 1.1



This XML file does not appear to have any style information associated with it. The document tree is shown below.

```
<ProductDetails>
  <Product>
    <ProductId>1</ProductId>
    <ProductName>Product 1</ProductName>
    <Price>10</Price>
  </Product>
</ProductDetails>
```



Product Details

Enter Product ID:

ProductId	ProductName	Price
1	Product 1	10

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2. Develop a database-driven WCF service to fetch and update user details.

User.cs

```
namespace UserWCFService
{
    using System;
    using System.Collections.Generic;
    public partial class User
    {
        public int UserId { get; set; }
        public string FirstName { get; set; }
        public string LastName { get; set; }
        public string Email { get; set; }
        public Nullable<int> Age { get; set; }
    }
}
```

IService1.cs

```
using System.ServiceModel;
namespace UserWCFService
{
    [ServiceContract]
    public interface IService1
    {
        [OperationContract]
        User GetUserDetails(int userId);
        [OperationContract]
        bool UpdateUserDetails(User user);
    }
}
```

Service1.svc

```
<%@ ServiceHost Language="C#" Debug="true"
Service="UserWCFService.Service1" CodeBehind="Service1.svc.cs" %>
```

Services1.svc.cs

```
using System; using System.Linq;
namespace UserWCFService
{
    public class Service1 : IService1
    {
        UserDBEntities db = new UserDBEntities();
        public User GetUserDetails(int userId)
```

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```
{
var user = db.Users.FirstOrDefault(u => u.UserId == userId);
if (user != null)
{
return new User
{
    UserId = user.UserId,
    FirstName = user.FirstName,
    LastName = user.LastName,
    Email = user.Email,
    Age = user.Age
};
}
return null;
}
public bool UpdateUserDetails(User user)
{
var existingUser = db.Users.FirstOrDefault(u => u.UserId == user.UserId);
if (existingUser != null)
{
existingUser.FirstName = user.FirstName;
existingUser.LastName = user.LastName;
existingUser.Email = user.Email;
existingUser.Age = user.Age;
db.SaveChanges();
return true;
}
return false;
}}}
```

Default.aspx

```
<%@ Page Language="C#" AutoEventWireup="true"
CodeBehind="Default.aspx.cs" Inherits="UserWebApp._Default" %>
<!DOCTYPE html>
<html>
<head runat="server">
<title>WCF User Details</title>
</head>
<body>
<form id="form1" runat="server">
<h2>WCF User Details Service</h2>
Enter User ID:
<asp:TextBox ID="txtUserId" runat="server"></asp:TextBox>
<asp:Button ID="btnFetch" runat="server" Text="Fetch User" OnClick="btnFetch_Click" />
<br /><br />
<asp:GridView ID="GridView1" runat="server"></asp:GridView>
<h3>Update User
Details</h3>
First Name:
```

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```
<asp:TextBox ID="txtFirstName" runat="server"></asp:TextBox><br />
<asp:TextBox ID="txtLastName" runat="server"></asp:TextBox><br />
<asp:TextBox ID="txtEmail" runat="server"></asp:TextBox><br />
Age:
<asp:TextBox ID="txtAge" runat="server"></asp:TextBox><br />
<asp:Button ID="btnUpdate" runat="server" Text="Update User"
OnClick="btnUpdate_Click" />
<br /><br />
<asp:Label ID="lblMessage" runat="server"
ForeColor="Green"></asp:Label>    </form>
</body> </html>
```

Last Name:

Email:

Default.aspx.cs

```
using System;
using System.Collections.Generic; using UserWebApp.ServiceReference1; namespace
UserWebApp
{
public partial class _Default : System.Web.UI.Page
{
Service1Client client = new Service1Client();
protected void btnFetch_Click(object sender, EventArgs e)
{
int id = Convert.ToInt32(txtUserId.Text);          var user = client.GetUserDetails(id);
if (user != null)
{
List<User> list = new List<User>();
list.Add(user);
GridView1.DataSource = list;
GridView1.DataBind();
txtFirstName.Text = user.FirstName;
txtLastName.Text = user.LastName;
txtEmail.Text = user.Email;
txtAge.Text = user.Age.ToString();
}}
protected void btnUpdate_Click(object sender, EventArgs e)
{
User user = new User();
user.UserId = Convert.ToInt32(txtUserId.Text);
user.FirstName = txtFirstName.Text;
user.LastName = txtLastName.Text;
user.Email = txtEmail.Text;
user.Age = Convert.ToInt32(txtAge.Text);
bool result = client.UpdateUserDetails(user);
if (result)
{
```


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```
lblMessage.Text = "User Updated Successfully!";  
}  
else  
{  
lblMessage.Text = "Update Failed!";  
}  
}}}
```

Database (UserDB)

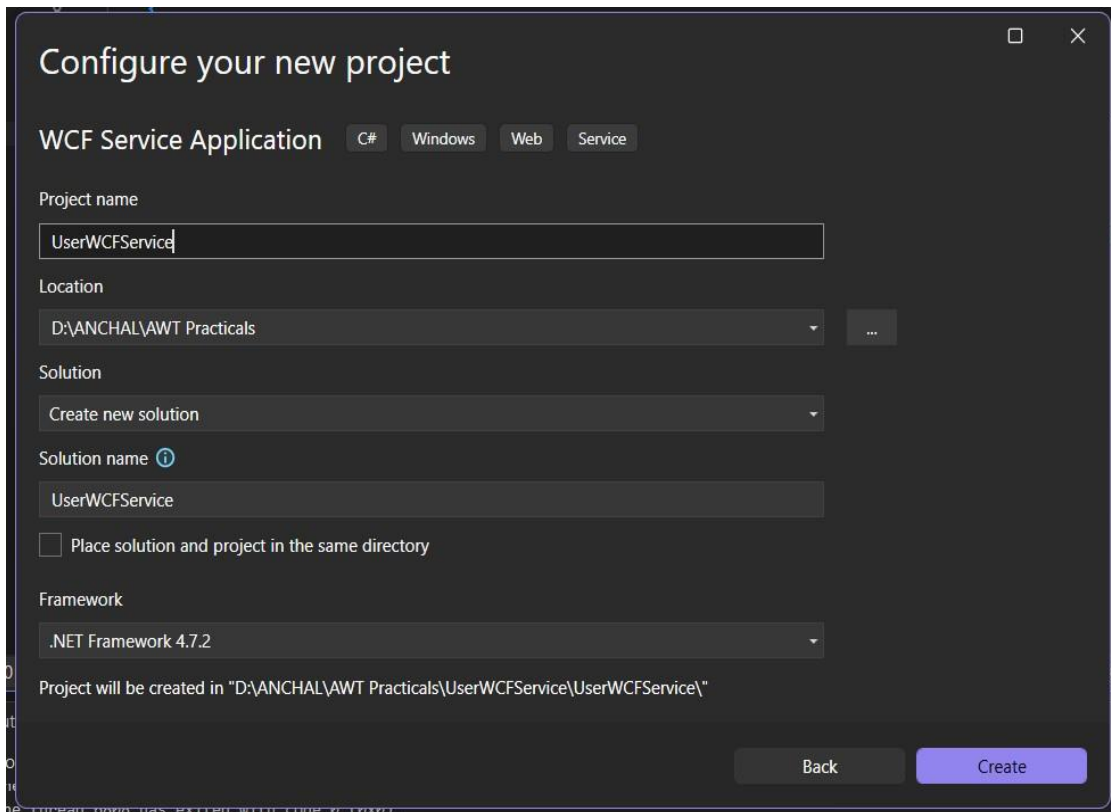
```
CREATE TABLE Users (  
  UserId INT PRIMARY KEY AUTO_INCREMENT,  
  FirstName VARCHAR(50),  
  LastName VARCHAR(50),  
  Email VARCHAR(100),  
  Age INT  
);
```

Output:

Results		Messages			
	UserId	FirstName	LastName	Email	Age
1	1	Anchal	Singh	anchal@gmail.com	22
2	2	Ravi	Kumar	ravi@gmail.com	25

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Configure your new project

WCF Service Application C# Windows Web Service

Project name
UserWCFService

Location
D:\ANCHAL\AWT Practicals

Solution
Create new solution

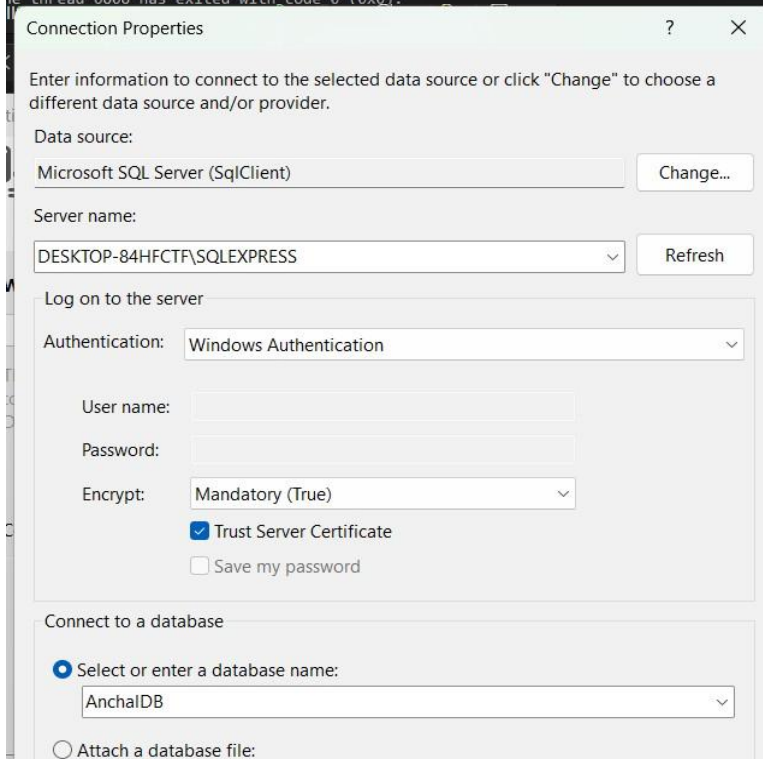
Solution name ⓘ
UserWCFService

☐ Place solution and project in the same directory

Framework
.NET Framework 4.7.2

Project will be created in "D:\ANCHAL\AWT Practicals\UserWCFService\UserWCFService\"

Back Create



Connection Properties

Enter information to connect to the selected data source or click "Change" to choose a different data source and/or provider.

Data source:
Microsoft SQL Server (SqlClient) Change...

Server name:
DESKTOP-84HFCTF\SQLEXPRESS Refresh

Log on to the server

Authentication: Windows Authentication

User name:

Password:

Encrypt: Mandatory (True)

☒ Trust Server Certificate

☐ Save my password

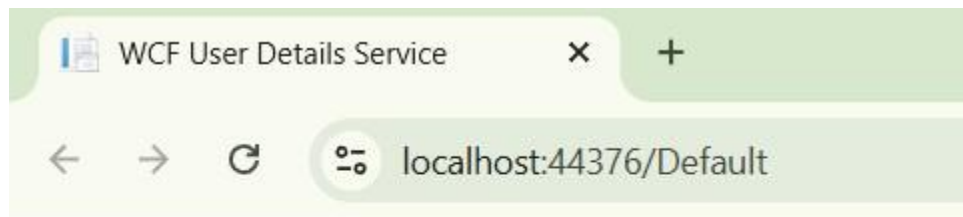
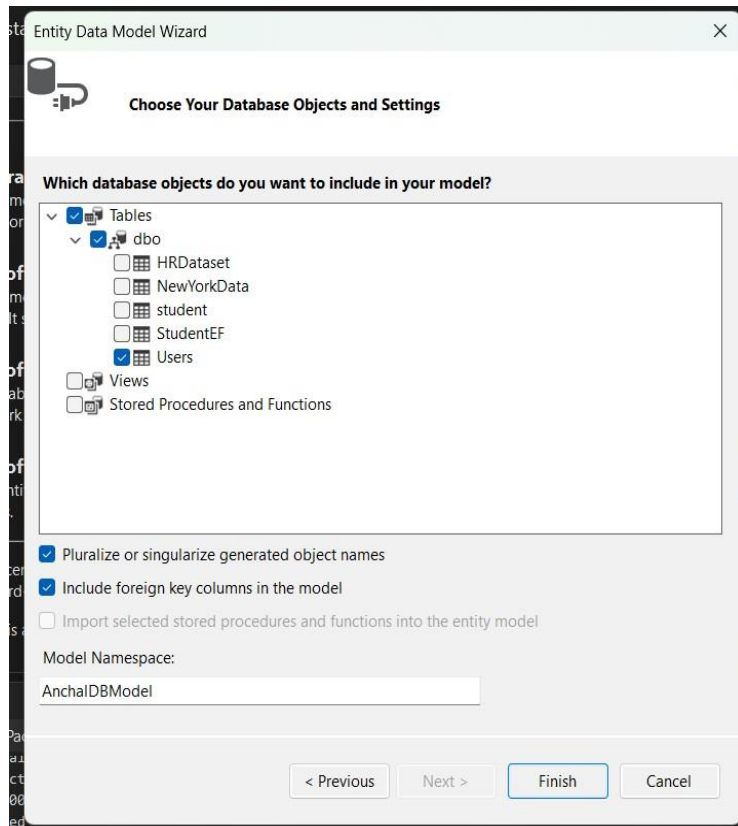
Connect to a database

☒ Select or enter a database name:
AnchalDB

☐ Attach a database file:

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WCF User Details Service

Enter User ID:

Age	Email	FirstName	LastName	UserId
22	anchal@gmail.com	Anchal	Singh	1

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3.Create xml based webservice to create calculator and consume it in website

Api:

CalculatorController.cs

```
using System;
using System.Net;
using System.Net.Http;
using System.Web.Http;
namespace CalculatorWebService.Controllers
{
    [RoutePrefix("api/calculator")]
    public class CalculatorController : ApiController
    {
        [HttpGet]
        [Route("add")]
        public HttpResponseMessage Add(int a, int b)
        {
            var result = a + b;
            return Request.CreateResponse(HttpStatusCode.OK, new { result });
        }

        [HttpGet]
        [Route("subtract")]
        public HttpResponseMessage Subtract(int a, int b)
        {
            var result = a - b;
            return Request.CreateResponse(HttpStatusCode.OK, new { result });
        }

        [HttpGet]
        [Route("multiply")]
        public HttpResponseMessage Multiply(int a, int b)
        {
            var result = a * b;
            return Request.CreateResponse(HttpStatusCode.OK, new { result });
        }

        [HttpGet]
        [Route("divide")]
        public HttpResponseMessage Divide(int a, int b)
        {
            if (b == 0)
                return Request.CreateErrorResponse(HttpStatusCode.BadRequest, "Cannot divide by zero.");
            var result = a / b;
            return Request.CreateResponse(HttpStatusCode.OK, new { result });
        }
    }
}
```

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```
}  
}  
}
```

Client:

Default.aspx

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="Default.aspx.cs"  
Inherits="CalculatorWebClient_119.Default" Async="true" %>  
<!DOCTYPE html>  
<html xmlns="http://www.w3.org/1999/xhtml">  
<head runat="server">  
<title>Calculator</title>  
</head>  
<body>  
<form id="form1" runat="server">  
<div>  
<h2>Calculator Web Service</h2>  
<h3>Anchal 119</h3>  
<br />  
A:  
<asp:TextBox ID="txtA" runat="server"></asp:TextBox>  
<br /><br />  
B:  
<asp:TextBox ID="txtB" runat="server"></asp:TextBox>  
<br /><br />  
<asp:Button ID="btnAdd" runat="server" Text="Add" OnClick="btnAdd_Click" /> <br  
</br />  
Result:  
<asp:TextBox ID="txtResult" runat="server" /></asp:TextBox>  
</div>  
</form>  
</body>  
</html>
```

Default.aspx.cs

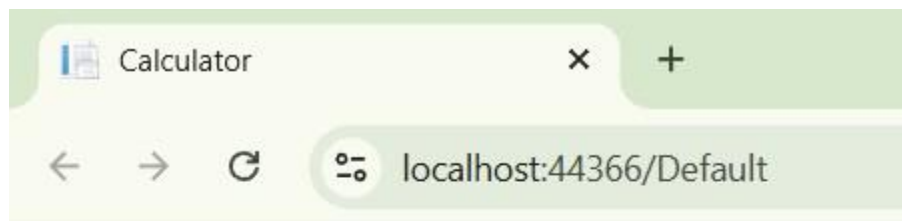
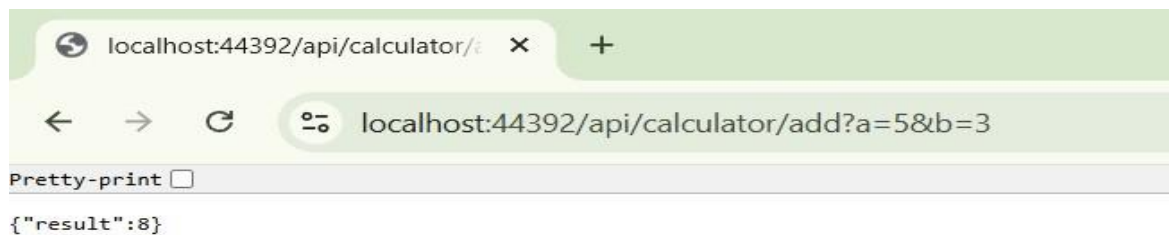
```
using System;  
using System.Net.Http;  
using System.Xml.Linq;  
namespace CalculatorWebClient_119  
{  
    public partial class Default : System.Web.UI.Page  
    {  
        protected async void btnAdd_Click(object sender, EventArgs e)
```

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```
{
int a = int.Parse(txtA.Text);
int b = int.Parse(txtB.Text);
using (HttpClient client = new HttpClient())
{
client.BaseAddress = new Uri("https://localhost:44392/api/calculator/"); // change port
HttpResponseMessage response = await client.GetAsync($"add?a={a}&b={b}");
if (response.IsSuccessStatusCode)
{
var xml = await response.Content.ReadAsStringAsync();
XElement data = XElement.Parse(xml);
txtResult.Text = data.Value;
}
else
{
txtResult.Text = "Error";}}}}}
```

Output:



Calculator Web Service

Anchal 119

A:

B:

Result:

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4. Create and consume the WCF service to calculate simple interest.

WCF Service Application -> WCF Service Application (.NET Framework)

Service1.svc.cs

```
using SimpleInterestWCF_119;
using System;
using System.Collections.Generic;
using System.Linq;
using System.Runtime.Serialization;
using System.ServiceModel;
using System.ServiceModel.Web;
using System.Text;
namespace SimpleInterestWCF_119
{
    public class Service1 : IService1
    {
        public double CalculateSimpleInterest(double principal, double rate, double time)
        {
            return (principal * rate * time) / 100;
        }
    }
}
```

IService1.cs

```
using System.ServiceModel;
namespace SimpleInterestWCF_119
{
    [ServiceContract]
    public interface IService1
    {
        [OperationContract]
        double CalculateSimpleInterest(double principal, double rate, double time);
    }
}
```

Web.config

```
<?xml version="1.0"?>
<configuration>
  <appSettings>
    <add key="aspnet:UseTaskFriendlySynchronizationContext" value="true" />
  
```

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```
</appSettings>
<system.web>
  <compilation debug="true" targetFramework="4.7.2" />
  <httpRuntime targetFramework="4.7.2" />
</system.web>
<system.serviceModel>
  <!-- BINDINGS -->
  <bindings>
    <basicHttpBinding>
      <binding name="BasicHttpBinding_IService1" />
    </basicHttpBinding>
  </bindings>
  <!-- SERVICES (VERY IMPORTANT) -->
  <services>
    <service name="SimpleInterestWCF_119.Service1">
      <!-- Actual service endpoint -->
      <endpoint
        address=""
        binding="basicHttpBinding"
        contract="SimpleInterestWCF_119.IService1" />
      <!-- Metadata exchange endpoint -->
      <endpoint
        address="mex"
        binding="mexHttpBinding"
        contract="IMetadataExchange" />
    </service>
  </services>
  <!-- BEHAVIORS -->
  <behaviors>
    <serviceBehaviors>
      <behavior>
        <serviceMetadata httpGetEnabled="true" httpsGetEnabled="true"/>
        <serviceDebug includeExceptionDetailInFaults="true"/>
      </behavior>
    </serviceBehaviors>
  </behaviors>
  <!-- HTTPS mapping (optional, safe to keep) -->
  <protocolMapping>
    <add binding="basicHttpsBinding" scheme="https" />
  </protocolMapping>
  <serviceHostingEnvironment
    aspNetCompatibilityEnabled="true"
    multipleSiteBindingsEnabled="true" />
</system.serviceModel>
<system.webServer>
  <modules runAllManagedModulesForAllRequests="true" />
  <directoryBrowse enabled="true" />
</system.webServer>
```


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</configuration>

Console Project

Program.cs

```
using System;
using System.Threading.Tasks;
using ServiceReference1;
namespace SimpleInterestClient119
{
    class Program
    {
        static async Task Main(string[] args)
        {
            Console.WriteLine("Anchal 119");
            Service1Client client = new Service1Client();
            Console.Write("Enter Principal: ");
            double principal = Convert.ToDouble(Console.ReadLine());
            Console.Write("Enter Rate: ");
            double rate = Convert.ToDouble(Console.ReadLine());
            Console.Write("Enter Time: ");
            double time = Convert.ToDouble(Console.ReadLine());
            double result = await client.CalculateSimpleInterestAsync(principal, rate, time);
            Console.WriteLine("\nSimple Interest = " + result);
            Console.ReadLine();
        }
    }
}
```

Output:

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Service1 Service

You have created a service.

To test this service, you will need to create a client and use it to call the service. You can do this using the svcutil.exe tool from the command line with the following syntax:

```
svcutil.exe http://localhost:54267/Service1.svc?wsdl
```

You can also access the service description as a single file:

```
http://localhost:54267/Service1.svc?singleWsdl
```

This will generate a configuration file and a code file that contains the client class. Add the two files to your client application and use the generated client class to call the Service. For example:

C#

```
class Test
{
    static void Main()
    {
        Service1Client client = new Service1Client();

        // Use the 'client' variable to call operations on the service.

        // Always close the client.
        client.Close();
    }
}
```

Visual Basic

```
Class Test
    Shared Sub Main()
        Dim client As Service1Client = New Service1Client()
        ' Use the 'client' variable to call operations on the service.

        ' Always close the client.
        client.Close()
    End Sub
End Class
```

Right-click Solution → Add → New Project → Console App (.NET Framework) Right-click Console Project → Add → Service Reference

Configure your new project

Console App C# Linux macOS Windows Console

Project name
SimpleInterestClient119

Location
D:\ANCHAL\AWT Practicals\SimpleInterestWCF_119

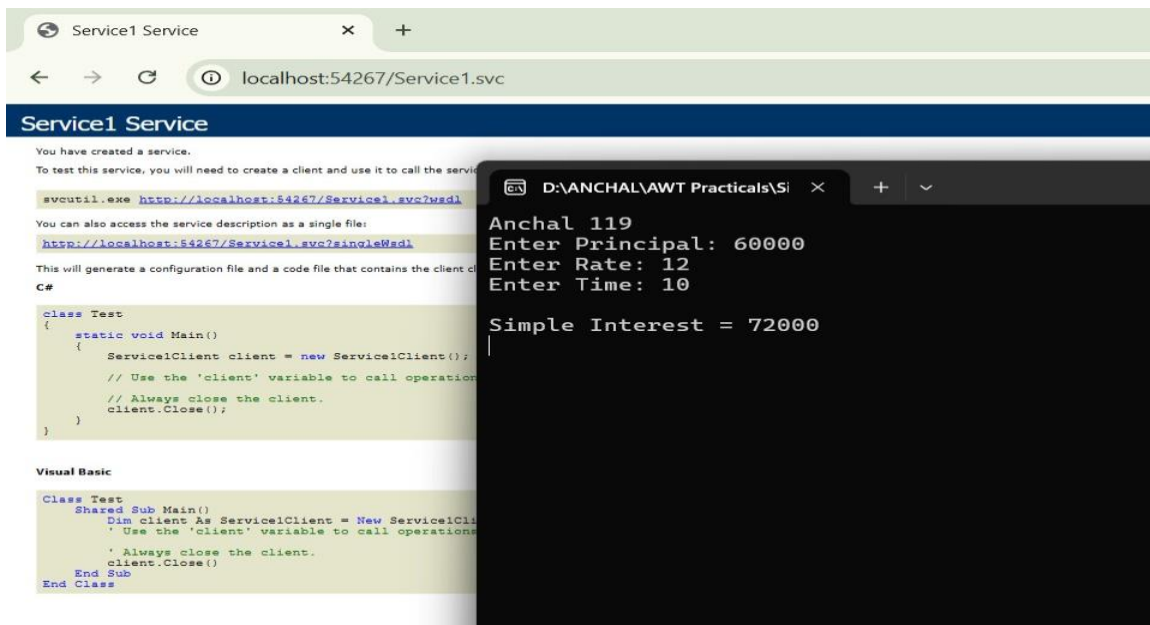
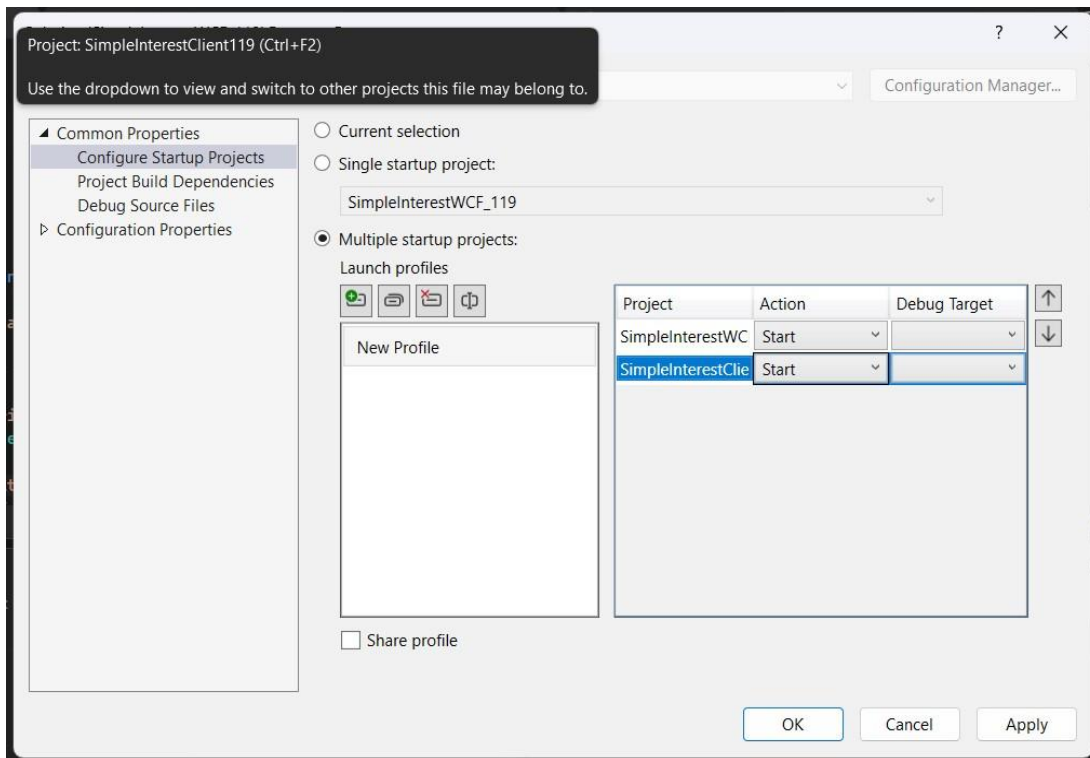
Project will be created in "D:\ANCHAL\AWT Practicals\SimpleInterestWCF_119\SimpleInterestClient119\"

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Right-click Solution -> Select Set Startup Projects -> Multiple startup projects



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Module 6: ASP.NET Core MVC Framework

1. Create an simple ASP.NET Core MVC application with models, views, and controllers to display book info

Step 1: Create a New ASP.NET Core MVC Project

Open Visual Studio or Visual Studio Code and create a new ASP.NET Core Web Application. Choose the ASP.NET Core Web App (Model-View-Controller) template. Make sure to select the latest version of .NET Core or .NET 5+ and click Create.

Step 2: Define the Model

You will define a Book model that contains properties like Title, Author, Genre, and Year.

1. In the Models folder, create a new class called

Book.cs

```
namespace BookStoreApp.Models
{
    public class Book
    {
        public int Id { get; set; }
        public string Title { get; set; }
        public string Author { get; set; }
        public string Genre { get; set; }
        public int Year { get; set; }
    }
}
```

Step 3: Create the Controller

Now, create a controller that will handle HTTP requests and pass data to the views.

1. In the Controllers folder, create a new controller named

BooksController.cs:

```
using Microsoft.AspNetCore.Mvc;
using BookStoreApp.Models;
using System.Collections.Generic;
namespace BookStoreApp.Controllers
{
    public class BooksController : Controller
    {
        public static List<Book> Books = new List<Book>
        {
```

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```
new Book { Id = 1, Title = "The Catcher in the Rye", Author = "J.D.
Salinger", Genre = "Fiction", Year = 1951 },
new Book { Id = 2, Title = "To Kill a Mockingbird", Author = "Harper Lee",
Genre = "Fiction", Year = 1960 },
new Book { Id = 3, Title = "1984", Author = "George Orwell", Genre =
"Dystopian", Year = 1949 },
new Book { Id = 4, Title = "Moby Dick", Author = "Herman Melville", Genre
= "Adventure", Year = 1851 }
};
```

```
public IActionResult Index()
{
    return View(Books);
}
```

```
public IActionResult Details(int id)
{
    var book = Books.Find(b => b.Id == id);    if (book == null)        return
    NotFound();

    return View(book);
}
}
```

Step 4: Create the Views

1. In the Views/Books folder, create two views: Index.cshtml and Details.cshtml.

Index.cshtml (Displays a list of books)

```
@model List<BookStoreApp.Models.Book>
<h1>Books List</h1>
<table class="table">
<thead>
<tr>
<th>Title</th>
<th>Author</th>
<th>Genre</th>
<th>Year</th>
<th>Actions</th>
</tr>
</thead>

<tbody>
@foreach (var book in Model)
{
```

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```
<tr>
<td>@book.Title</td>
<td>@book.Author</td>
<td>@book.Genre</td>
<td>@book.Year</td>
<td>
<a href="@Url.Action("Details", "Books", new { id = book.Id })"
btn-info">Details</a>
class="btn
</td>
</tr>
}
</tbody>
</table>
```

Details.cshtml (Displays details of a single book)

```
@model BookStoreApp.Models.Book
<h1>Book Details</h1>
<div>
<h3>@Model.Title</h3>
<p><b>Author:</b> @Model.Author</p>
<p><b>Genre:</b> @Model.Genre</p>
<p><b>Year:</b> @Model.Year</p>
<a href="@Url.Action("Index", "Books")" class="btn btn-primary">
Back to List
</a>
</div>
```

Update the Program.cs

```
var builder = WebApplication.CreateBuilder(args);
// Add services to the container.
builder.Services.AddControllersWithViews();
var app = builder.Build();
// Configure the HTTP request pipeline. if (!app.Environment.IsDevelopment())
{
app.UseExceptionHandler("/Home/Error");
app.UseHsts();
}
app.UseHttpsRedirection(); app.UseStaticFiles();
app.UseRouting();
app.UseAuthorization();
app.MapControllerRoute( name: "default",
pattern: "{controller=Books}/{action=Index}/{id?}");
app.Run();
```

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Output:

The screenshot shows a web browser window with the address bar at `localhost:7121`. The page title is "BookStoreApp" and it has navigation links for "Home" and "Privacy". The main content area is titled "Books List" and contains a table with the following data:

Title	Author	Genre	Year	Actions
The Catcher in the Rye	J.D. Salinger	Fiction	1951	Details
To Kill a Mockingbird	Harper Lee	Fiction	1960	Details
1984	George Orwell	Dystopian	1949	Details
Moby Dick	Herman Melville	Adventure	1851	Details

Below the browser window, a terminal window shows the application's startup logs:

```
info: Microsoft.Hosting.Lifetime[14]
      Now listening on: https://localhost:7121
info: Microsoft.Hosting.Lifetime[14]
      Now listening on: http://localhost:5216
info: Microsoft.Hosting.Lifetime[0]
      Application started. Press Ctrl+C to shut down.
info: Microsoft.Hosting.Lifetime[0]
      Hosting environment: Development
info: Microsoft.Hosting.Lifetime[0]
      Content root path: D:\ANCHAL\AWT Practicals\BookStoreApp\BookStoreApp
```

The screenshot shows a web browser window with the address bar at `localhost:7121/Books/Details/1`. The page title is "BookStoreApp" and it has navigation links for "Home" and "Privacy". The main content area is titled "Book Details" and displays the following information:

The Catcher in the Rye
Author: J.D. Salinger
Genre: Fiction
Year: 1951

At the bottom of the page, there is a blue button labeled "Back to List".

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2. Design a form with data annotations for validation and display appropriate error messages.

ASP.NET Core Web App (Model-View-Controller) -> UserFormValidation

Create the Model

UserViewModel.cs

```
using System.ComponentModel.DataAnnotations;
public class UserViewModel
{
    [Required(ErrorMessage = "Name is required")]
    [StringLength(50, ErrorMessage = "Name cannot be longer than 50 characters")]    public
    string Name { get; set; }
    [Required(ErrorMessage = "Email is required")]
    [EmailAddress(ErrorMessage = "Invalid email address")]    public string Email { get; set; }
    [Range(18, 100, ErrorMessage = "Age must be between 18 and 100")]    public int? Age {
    get; set; }
}
```

Create the Controller UserController.cs

```
using Microsoft.AspNetCore.Mvc;
public class UserController : Controller
{
    [HttpGet]
    public IActionResult Register()
    {
        return View();
    }
    [HttpPost]
    public IActionResult Register(UserViewModel model)
    {
        if (ModelState.IsValid)
        {
            return RedirectToAction("Success");
        }
        return View(model);
    }
    public IActionResult Success()
    {
        return View();
    }
}
```


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}

Create the View (Register.cshtml)

Views folder → Add → New Folder → Name: User

User folder → Add → View -> Register.cshtml

```
@model UserViewModel
@{
    ViewData["Title"] = "User Registration";
}
<h2>User Registration</h2>
<form asp-action="Register" method="post">
    <div class="form-group">
        <label asp-for="Name"></label>
        <input asp-for="Name" class="form-control" />
        <span asp-validation-for="Name" class="text-danger"></span>
    </div>
    <div class="form-group">
        <label asp-for="Email"></label>
        <input asp-for="Email" class="form-control" />
        <span asp-validation-for="Email" class="text-danger"></span>
    </div>
    <div class="form-group">
        <label asp-for="Age"></label>
        <input asp-for="Age" class="form-control" />
        <span asp-validation-for="Age" class="text-danger"></span>
    </div>
    <button type="submit" class="btn btn-primary">Register</button>
</form>
@section Scripts {
    @{
        await Html.RenderPartialAsync("_ValidationScriptsPartial");
    }
}
```

Success.cshtml

```
<h2>Registration Successful!</h2>
<p>Your form was submitted successfully.</p>
```

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User Registration - UserFormVa x +

localhost:7299/User/Register

UserFormValidation Home Privacy

User Registration

Name

Anchal

Email

anchal@gmail.com

Age

23

Register

- UserFormValidation x +

localhost:7299/User/Success

UserFormValidation Home Privacy

Registration Successful!

Your form was submitted successfully.

User Registration - UserFormVa x +

localhost:7299/User/Register

UserFormValidation Home Privacy

User Registration

Name

Anchal

Email

anchalgmail.com

Invalid email address

Age

23

Register

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3. Implement CRUD operations using Entity Framework Code-First approach for emp data.

Create ASP.NET Core Web API Project

In Visual Studio or using CLI: **dotnet new webapi -n EmpCRUD cd EmpCRUD**

```
C:\Windows\System32\cmd.e... X + v
Microsoft Windows [Version 10.0.26200.8328]
(c) Microsoft Corporation. All rights reserved.

D:\ANCHAL\AWT Practicals>dotnet tool install --global dotnet-ef
Tool 'dotnet-ef' is already installed.

D:\ANCHAL\AWT Practicals>dotnet new webapi -n EmpCRUD
The template "ASP.NET Core Web API" was created successfully.

Processing post-creation actions...
Restoring D:\ANCHAL\AWT Practicals\EmpCRUD\EmpCRUD.csproj:
Restore succeeded.
```

Install EF Core NuGet Packages dotnet add package

Microsoft.EntityFrameworkCore.SqlServer dotnet add package

Microsoft.EntityFrameworkCore.Tools

```
D:\ANCHAL\AWT Practicals>cd EmpCRUD

D:\ANCHAL\AWT Practicals\EmpCRUD>dotnet add package Microsoft.EntityFrameworkCore.SqlServer
info : X.509 certificate chain validation will use the default trust store selected by .NET for code signing.
info : X.509 certificate chain validation will use the default trust store selected by .NET for timestamping.
info : Adding PackageReference for package 'Microsoft.EntityFrameworkCore.SqlServer' into project 'D:\ANCHAL\AWT Practicals\EmpCRUD\EmpCRUD.csproj'.
info :   GET https://api.nuget.org/v3/registration5-gz-semver2/microsoft.entityframeworkcore.sqlserver/index.json
info :   OK https://api.nuget.org/v3/registration5-gz-semver2/microsoft.entityframeworkcore.sqlserver/index.json 1029ms
info :   GET https://api.nuget.org/v3/registration5-gz-semver2/microsoft.entityframeworkcore.sqlserver/page/0.0.1-alpha/3.1.2.json
info :   OK https://api.nuget.org/v3/registration5-gz-semver2/microsoft.entityframeworkcore.sqlserver/page/0.0.1-alpha/3.1.2.json 814ms

D:\ANCHAL\AWT Practicals\EmpCRUD>dotnet add package Microsoft.EntityFrameworkCore.Tools
info : X.509 certificate chain validation will use the default trust store selected by .NET for code signing.
info : X.509 certificate chain validation will use the default trust store selected by .NET for timestamping.
info : Adding PackageReference for package 'Microsoft.EntityFrameworkCore.Tools' into project 'D:\ANCHAL\AWT Practicals\EmpCRUD\EmpCRUD.csproj'.
info :   GET https://api.nuget.org/v3/registration5-gz-semver2/microsoft.entityframeworkcore.tools/index.json
info :   OK https://api.nuget.org/v3/registration5-gz-semver2/microsoft.entityframeworkcore.tools/index.json 4894ms
info :   GET https://api.nuget.org/v3/registration5-gz-semver2/microsoft.entityframeworkcore.tools/page/1.0.0-msbuild1-final/3.1.0.json
info :   OK https://api.nuget.org/v3/registration5-gz-semver2/microsoft.entityframeworkcore.tools/page/1.0.0-msbuild1-final/3.1.0.json 814ms
info :   GET https://api.nuget.org/v3/registration5-gz-semver2/microsoft.entityframeworkcore.tools/page/3.1.1/6.0.0-preview.4.21253.1.json
info :   OK https://api.nuget.org/v3/registration5-gz-semver2/microsoft.entityframeworkcore.tools/page/3.1.1/6.0.0-preview.4.21253.1.json 814ms
```

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Creating the Employee Model

Models/Emp.cs

```
using System.ComponentModel.DataAnnotations;
namespace EmpCRUD.Models
{
    public class Emp
    {
        [Key]
        public int Id { get; set; }
        [Required]
        public string Name { get; set; }
        public string Department { get; set; }
        public decimal Salary { get; set; }
    }
}
```

Create the DbContext Data/EmpContext.cs

```
using Microsoft.EntityFrameworkCore; using EmpCRUD.Models;
namespace EmpCRUD.Data
{
    public class EmpContext : DbContext
    {
        public EmpContext(DbContextOptions<EmpContext> options) : base(options)
        {
        }
        public DbSet<Emp> Emps { get; set; }
    }
}
```

Configure the Database Connection

appsettings.json

```
{
  "ConnectionStrings": {
    "DefaultConnection":
    "Server=DESKTOP-
84HFCTF\\SQLEXPRESS;Database=AnchalDB;Trusted_Connection=True;TrustS
erverCertificate=True;"
  },
}
```

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```
"Logging": {  
  "LogLevel": {  
    "Default": "Information",  
    "Microsoft.AspNetCore": "Warning"  
  }  
},  
"AllowedHosts": "*" }  
}
```

Program.cs

```
using EmpCRUD.Data;  
using Microsoft.EntityFrameworkCore;  
var builder = WebApplication.CreateBuilder(args);  
// Add services to the container builder.Services.AddControllers();  
// Register EF Core + SQL Server builder.Services.AddDbContext<EmpContext>(options =>  
options.UseSqlServer(builder.Configuration.GetConnectionString("DefaultConnection")));  
// Optional: Swagger/OpenAPI builder.Services.AddEndpointsApiExplorer();  
builder.Services.AddSwaggerGen();  
var app = builder.Build();  
// Configure HTTP request pipeline if (app.Environment.IsDevelopment())  
{  
  app.UseSwagger();   app.UseSwaggerUI();  
}  
app.UseHttpsRedirection();  
app.MapControllers(); // <-- Important: Maps EmpController  
// (Optional) keep weather API var summaries = new[]  
{  
  "Freezing", "Bracing", "Chilly", "Cool", "Mild",  
  "Warm", "Balmy", "Hot", "Sweltering", "Scorching"  
};  
  
app.MapGet("/weatherforecast", () =>  
{  
  var forecast = Enumerable.Range(1, 5).Select(index =>      new WeatherForecast  
(  
    DateOnly.FromDateTime(DateTime.Now.AddDays(index)),      Random.Shared.Next(-20,  
55),      summaries[Random.Shared.Next(summaries.Length)]  
))  
  .ToArray();  
  
  return forecast;  
})  
.WithName("GetWeatherForecast");
```

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```
app.Run();
record WeatherForecast(DateOnly Date, int TemperatureC, string? Summary)
{
    public int TemperatureF => 32 + (int)(TemperatureC / 0.5556);
}
```

Add Migration and Update Database

dotnet ef migrations add Initial
Create dotnet ef database update

```
D:\ANCHAL\AWT Practicals\EmpCRUD>dotnet ef migrations add InitialCreate
Build started...
Build succeeded.
No store type was specified for the decimal property 'Salary' on entity type 'Emp'. This will cause values to be silently truncated if they do not fit in the default precision and scale. Explicitly specify the SQL server column type that can accommodate all the values in 'OnModelCreating' using 'HasColumnType', specify precision and scale using 'HasPrecision', or configure a value converter using 'HasConversion'.
Done. To undo this action, use 'ef migrations remove'

D:\ANCHAL\AWT Practicals\EmpCRUD>dotnet ef database update
Build started...
Build succeeded.
Applying migration '20260505040203_InitialCreate'.
Done.
```

Create the API Controller

Controllers/EmpController.cs

```
using EmpCRUD.Data;
using EmpCRUD.Models;
using Microsoft.AspNetCore.Mvc;
using Microsoft.EntityFrameworkCore;
namespace EmpCRUD.Controllers
{
    [Route("api/[controller]")]
    [ApiController] public class EmpController : ControllerBase
    {
        private readonly EmpContext _context;

        public EmpController(EmpContext context)
        {
            _context = context;
        }
        // GET: api/emp
        [HttpGet]
        public async Task<ActionResult<IEnumerable<Emp>>> GetEmps()
        {
            return await _context.Emps.ToListAsync();
        }
    }
}
```

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```
// GET: api/emp/5      [HttpGet("{id}")]      public async
Task<ActionResult<Emp>>GetEmp(int id)
{
    var emp = await _context.Emps.FindAsync(id);
    if (emp == null)
    {
        return NotFound();
    }
    return emp;
}
// POST: api/emp
[HttpPost]
public async Task<ActionResult<Emp>>PostEmp(Emp emp)
{
    _context.Emps.Add(emp);          await _context.SaveChangesAsync();          return
    CreatedAtAction(nameof(GetEmp), new { id = emp.Id }, emp);
}
// PUT: api/emp/5
[HttpPut("{id}")]
public async Task<IActionResult>PutEmp(int id, Emp emp)
{
    if (id != emp.Id)
    {
        return BadRequest();
    }
    _context.Entry(emp).State = EntityState.Modified;
    try
    {
        await _context.SaveChangesAsync();
    }
    catch (DbUpdateConcurrencyException)
    {
        if (!_context.Emps.Any(e => e.Id == id))
        {
            return NotFound();
        }
        throw;
    }
    return NoContent();
}
// DELETE: api/emp/5      [HttpDelete("{id}")]      public async
Task<IActionResult>DeleteEmp(int id)
{
    var emp = await _context.Emps.FindAsync(id);
    if (emp == null)
    {
```

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```
return NotFound();
}
_context.Emps.Remove(emp);
await _context.SaveChangesAsync();
return NoContent();
}}}
```

Output:

```
D:\ANCHAL\AWT Practicals\EmpCRUD>dotnet add package Swashbuckle.AspNetCore
info : X.509 certificate chain validation will use the default trust store selected by .NET for code signing.
info : X.509 certificate chain validation will use the default trust store selected by .NET for timestamping.
info : Adding PackageReference for package 'Swashbuckle.AspNetCore' into project 'D:\ANCHAL\AWT Practicals\EmpCRUD\EmpCRUD.csproj'.
info : GET https://api.nuget.org/v3/registration5-gz-semver2/swashbuckle.aspnetcore/index.json
info : OK https://api.nuget.org/v3/registration5-gz-semver2/swashbuckle.aspnetcore/index.json 441ms
info : Restoring packages for D:\ANCHAL\AWT Practicals\EmpCRUD\EmpCRUD.csproj...
info : GET https://api.nuget.org/v3-flatcontainer/swashbuckle.aspnetcore/index.json
info : OK https://api.nuget.org/v3-flatcontainer/swashbuckle.aspnetcore/index.json 37ms

D:\ANCHAL\AWT Practicals\EmpCRUD>dotnet ef migrations add InitialCreate
Build started...
Build succeeded.
No store type was specified for the decimal property 'Salary' on entity type 'Emp'. This will cause values to be silently truncated if they do not fit in the default precision and scale. Explicitly specify the SQL server column type that can accommodate all the values in 'OnModelCreating' using 'HasColumnType', specify precision and scale using 'HasPrecision', or configure a value converter using 'HasConversion'.
Done. To undo this action, use 'ef migrations remove'

D:\ANCHAL\AWT Practicals\EmpCRUD>dotnet ef database update
Build started...
Build succeeded.
Applying migration '20260505040203_InitialCreate'.
Done.
```

```
D:\ANCHAL\AWT Practicals\EmpCRUD>dotnet build
Restore complete (1.5s)
info NETSDK1057: You are using a preview version of .NET. See: https://aka.ms/dotnet-support-policy
EmpCRUD net10.0 succeeded with 2 warning(s) (7.6s) → bin\Debug\net10.0\EmpCRUD.dll
D:\ANCHAL\AWT Practicals\EmpCRUD\Models\Emp.cs(11,23): warning CS8618: Non-nullable property 'Name' must contain a non-null value when exiting constructor. Consider adding the 'required' modifier or declaring the property as nullable.
D:\ANCHAL\AWT Practicals\EmpCRUD\Models\Emp.cs(13,23): warning CS8618: Non-nullable property 'Department' must contain a non-null value when exiting constructor. Consider adding the 'required' modifier or declaring the property as nullable.

Build succeeded with 2 warning(s) in 10.6s
```

```
D:\ANCHAL\AWT Practicals\EmpCRUD>dotnet run
Using launch settings from D:\ANCHAL\AWT Practicals\EmpCRUD\Properties\launchSettings.json...
Building...
Restore succeeded with 4 warning(s) in 1.1s
D:\ANCHAL\AWT Practicals\EmpCRUD\EmpCRUD.csproj : warning NU1902: Package 'Azure.Identity' 1.10.3 has a known moderate severity vulnerability, https://github.com/advisories/GHSA-m5vv-6r4h-3vj9
D:\ANCHAL\AWT Practicals\EmpCRUD\EmpCRUD.csproj : warning NU1902: Package 'Azure.Identity' 1.10.3 has a known moderate severity vulnerability, https://github.com/advisories/GHSA-wvxc-855f-jvrv
D:\ANCHAL\AWT Practicals\EmpCRUD\EmpCRUD.csproj : warning NU1902: Package 'Microsoft.Identity.Client' 4.56.0 has a known moderate severity vulnerability, https://github.com/advisories/GHSA-m5vv-6r4h-3vj9
D:\ANCHAL\AWT Practicals\EmpCRUD\EmpCRUD.csproj : warning NU1901: Package 'Microsoft.Identity.Client' 4.56.0 has a known low severity vulnerability, https://github.com/advisories/GHSA-x674-v45j-fwxw
EmpCRUD net10.0 succeeded with 4 warning(s) (0.9s) → bin\Debug\net10.0\EmpCRUD.dll
D:\ANCHAL\AWT Practicals\EmpCRUD\EmpCRUD.csproj : warning NU1902: Package 'Azure.Identity' 1.10.3 has a known moderate severity vulnerability, https://github.com/advisories/GHSA-m5vv-6r4h-3vj9
D:\ANCHAL\AWT Practicals\EmpCRUD\EmpCRUD.csproj : warning NU1902: Package 'Azure.Identity' 1.10.3 has a known moderate severity vulnerability, https://github.com/advisories/GHSA-wvxc-855f-jvrv
D:\ANCHAL\AWT Practicals\EmpCRUD\EmpCRUD.csproj : warning NU1902: Package 'Microsoft.Identity.Client' 4.56.0 has a known moderate severity vulnerability, https://github.com/advisories/GHSA-m5vv-6r4h-3vj9
D:\ANCHAL\AWT Practicals\EmpCRUD\EmpCRUD.csproj : warning NU1901: Package 'Microsoft.Identity.Client' 4.56.0 has a known low severity vulnerability, https://github.com/advisories/GHSA-x674-v45j-fwxw
info: Microsoft.Hosting.Lifetime[14]
      Now listening on: http://localhost:5001
info: Microsoft.Hosting.Lifetime[0]
      Application started. Press Ctrl+C to shut down.
```


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Browser tabs: AWT LAB - Google, Module 6 AWT 11, EF Core Warning, localhost:5001/ap, localhost:5001/W, Module 6 - Classi, Swagger UI

Address bar: localhost:5001/swagger/index.html

Swagger UI Header: Select a definition EmpCRUD v1

EmpCRUD 1.0 HAS 3.0

http://localhost:5001/swagger/v1/swagger.json

Emp

- GET /api/Emp
- POST /api/Emp
- GET /api/Emp/{id}
- PUT /api/Emp/{id}
- DELETE /api/Emp/{id}

EmpCRUD

- GET /weather-forecast

Schemas

EmpCRUD

GET /weather-forecast

Parameters

No parameters

Try it out

Responses

Code	Description	Links
200	OK	No links

Media type: application/json

Controls Accept header.

Example Value | Schema

```
{  "date": "2026-05-05",  "temperatureC": 0,  "summary": "string",  "temperatureF": 0}
```

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4. ASP.NET Core MVC with EF Core DB First Approach

Program.cs

```
using EmpMvcDbFirst.Models;
using Microsoft.EntityFrameworkCore;
var builder = WebApplication.CreateBuilder(args);
builder.Services.AddControllersWithViews();
builder.Services.AddDbContext<AnchalDbContext>(options =>
options.UseSqlServer(builder.Configuration.GetConnectionString("DefaultConnection")));
var app = builder.Build();
app.UseStaticFiles(); app.UseRouting();
app.MapControllerRoute( name: "default", pattern:
"{controller=Employees}/{action=Index}/{id?}");
app.Run();
```

Appsettings.json

```
{
  "ConnectionStrings": {    "DefaultConnection":
  "Server=DESKTOP-
84HFCTF\\SQLEXPRESS;Database=AnchalDB;Trusted_Connection=True;TrustServerCertif
  icat e=True;"
  },
  "Logging": {
    "LogLevel": {
      "Default": "Information",
      "Microsoft.AspNetCore": "Warning"
    }
  },
  "AllowedHosts": "*"
}
Index.cshtml
@model IEnumerable<EmpMvcDbFirst.Models.Employee>

@{
  ViewData["Title"] = "Employees";
}

<h2>Employees</h2>
<p><a asp-action="Create" class="btn btn-primary">Add New</a></p>

<table class="table table-bordered">
```

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```
<thead>
<tr>
<th>Id</th>
<th>Name</th>
<th>Email</th>
<th>Salary</th>
<th>Actions</th>
</tr>
</thead>
<tbody>
@foreach (var emp in Model)
{
<tr>
<td>@emp.Id</td>
<td>@emp.Name</td>
<td>@emp.Email</td>
<td>@emp.Salary</td>
<td>
<a asp-action="Edit" asp-route-id="@emp.Id" class="btn btn-warning btn-sm">Edit</a>
<a asp-action="Details" asp-route-id="@emp.Id" class="btn btn-info btn-sm">Details</a>
<a asp-action="Delete" asp-route-id="@emp.Id" class="btn btn-danger btn-sm">Delete</a>
</td>
</tr>
}
</tbody>
</table>
```

Create.cshtml

@model EmpMvcDbFirst.Models.Employee

```
@{
ViewData["Title"] = "Create Employee";
}
<h2>Create Employee</h2>
```

```
<form asp-action="Create">
<div asp-validation-summary="ModelOnly" class="text-danger"></div>
<div class="mb-3">
<label asp-for="Name" class="form-label"></label>
<input asp-for="Name" class="form-control" />
<span asp-validation-for="Name" class="text-danger"></span>
</div>
<div class="mb-3">
<label asp-for="Email" class="form-label"></label>
<input asp-for="Email" class="form-control" />
```

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```
<span asp-validation-for="Email" class="text-danger"></span>
</div>
<div class="mb-3">
<label asp-for="Salary" class="form-label"></label>
<input asp-for="Salary" class="form-control" />
<span asp-validation-for="Salary" class="text-danger"></span>
</div>

<button type="submit" class="btn btn-success">Create</button>
<a asp-action="Index" class="btn btn-secondary">Back</a>
</form>
```

```
@section Scripts {
@{
await Html.RenderPartialAsync("_ValidationScriptsPartial");
}
}
```

Details.cshtml

```
@model EmpMvcDbFirst.Models.Employee
```

```
@{
 ViewData["Title"] = "Employee Details";
}
```

```
<h2>Employee Details</h2>
```

```
<dl class="row">
<dt class="col-sm-2">Id</dt>
<dd class="col-sm-10">@Model.Id</dd>
```

```
<dt class="col-sm-2">Name</dt>
<dd class="col-sm-10">@Model.Name</dd>
```

```
<dt class="col-sm-2">Email</dt>
<dd class="col-sm-10">@Model.Email</dd>
```

```
<dt class="col-sm-2">Salary</dt>
<dd class="col-sm-10">@Model.Salary</dd>
</dl>
```

```
<a asp-action="Edit" asp-route-id="@Model.Id" class="btn btn-warning">Edit</a>
<a asp-action="Index" class="btn btn-secondary">Back</a>
```

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Edit.cshtml

```
@model EmpMvcDbFirst.Models.Employee
```

```
@{  
    ViewData["Title"] = "Edit Employee";  
}
```

```
<h2>Edit Employee</h2>
```

```
<form asp-action="Edit">  
    <div asp-validation-summary="ModelOnly" class="text-danger"></div>  
    <input type="hidden" asp-for="Id" />
```

```
<div class="mb-3">  
    <label asp-for="Name" class="form-label"></label>  
    <input asp-for="Name" class="form-control" />  
    <span asp-validation-for="Name" class="text-danger"></span>  
</div>
```

```
<div class="mb-3">  
    <label asp-for="Email" class="form-label"></label>  
    <input asp-for="Email" class="form-control" />  
    <span asp-validation-for="Email" class="text-danger"></span>  
</div>
```

```
<div class="mb-3">  
    <label asp-for="Salary" class="form-label"></label>  
    <input asp-for="Salary" class="form-control" />  
    <span asp-validation-for="Salary" class="text-danger"></span>  
</div>
```

```
<button type="submit" class="btn btn-warning">Save</button>  
<a asp-action="Index" class="btn btn-secondary">Back</a>  
</form>
```

```
@section Scripts {  
    @{  
        await Html.RenderPartialAsync("_ValidationScriptsPartial");  
    }  
}
```

Delete.cshtml

```
@model EmpMvcDbFirst.Models.Employee
```

```
@{  
    ViewData["Title"] = "Delete Employee";  
}
```

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<h2>Delete Employee</h2>

<p class="text-danger">Are you sure you want to delete this employee?</p>

<dl class="row">

<dt class="col-sm-2">Id</dt>

<dd class="col-sm-10">@Model.Id</dd>

<dt class="col-sm-2">Name</dt>

<dd class="col-sm-10">@Model.Name</dd>

<dt class="col-sm-2">Email</dt>

<dd class="col-sm-10">@Model.Email</dd>

<dt class="col-sm-2">Salary</dt>

<dd class="col-sm-10">@Model.Salary</dd>

</dl>

<form asp-action="Delete">

<input type="hidden" asp-for="Id" />

<button type="submit" class="btn btn-danger">Yes, Delete</button>

<a asp-action="Index" class="btn btn-secondary">Cancel

</form>

Output:

```
Command Prompt
Microsoft Windows [Version 10.0.26200.8328]
(c) Microsoft Corporation. All rights reserved.

C:\Users\MSI>dotnet tool install --global dotnet-ef
You can invoke the tool using the following command: dotnet-ef
Tool 'dotnet-ef' (version '10.0.7') was successfully installed.

C:\Users\MSI>
```

```
PM> Scaffold-DbContext "Server=DESKTOP-84HFCTF
\SQLEXPRESS;Database=AnchalDB;Trusted_Connection=True;TrustServerCertificate=True;" Microsoft.EntityFrameworkCore.SqlServer
-OutputDir Models
Build started...
Build succeeded.
To protect potentially sensitive information in your connection string, you should move it out of source code. You can
avoid scaffolding the connection string by using the Name= syntax to read it from configuration - see https://
go.microsoft.com/fwlink/?linkid=2131148. For more guidance on storing connection strings, see https://go.microsoft.com/
/fwlink/?LinkId=723263.
PM>
```

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```
PM> dotnet tool install -g dotnet-aspnet-codegenerator
You can invoke the tool using the following command: dotnet-aspnet-codegenerator
Tool 'dotnet-aspnet-codegenerator' (version '10.0.2') was successfully installed.
PM> |
```

```
PM> Install-Package Microsoft.VisualStudio.Web.CodeGeneration.Design -Version 8.0.7
Restoring packages for D:\ANCHAL\AWT Practicals\EmpMvcDbFirst\EmpMvcDbFirst\EmpMvcDbFirst.csproj...
GET https://api.nuget.org/v3-flatcontainer/microsoft.visualstudio.web.codegeneration.design/index.json
OK https://api.nuget.org/v3-flatcontainer/microsoft.visualstudio.web.codegeneration.design/index.json 330ms
GET https://api.nuget.org/v3-flatcontainer/microsoft.visualstudio.web.codegeneration.design/8.0.7/
microsoft.visualstudio.web.codegeneration.design.8.0.7.nupkg
OK https://api.nuget.org/v3-flatcontainer/microsoft.visualstudio.web.codegeneration.design/8.0.7/
microsoft.visualstudio.web.codegeneration.design.8.0.7.nupkg 22ms
GET https://api.nuget.org/v3-flatcontainer/microsoft.visualstudio.web.codegenerators.mvc/index.json
GET https://api.nuget.org/v3-flatcontainer/microsoft.dotnet.scaffolding.shared/index.json
OK https://api.nuget.org/v3-flatcontainer/microsoft.visualstudio.web.codegenerators.mvc/index.json 219ms
GET https://api.nuget.org/v3-flatcontainer/microsoft.visualstudio.web.codegenerators.mvc/8.0.7/
100 % ▾
```

Developer PowerShell

Error List

Package Manager Console

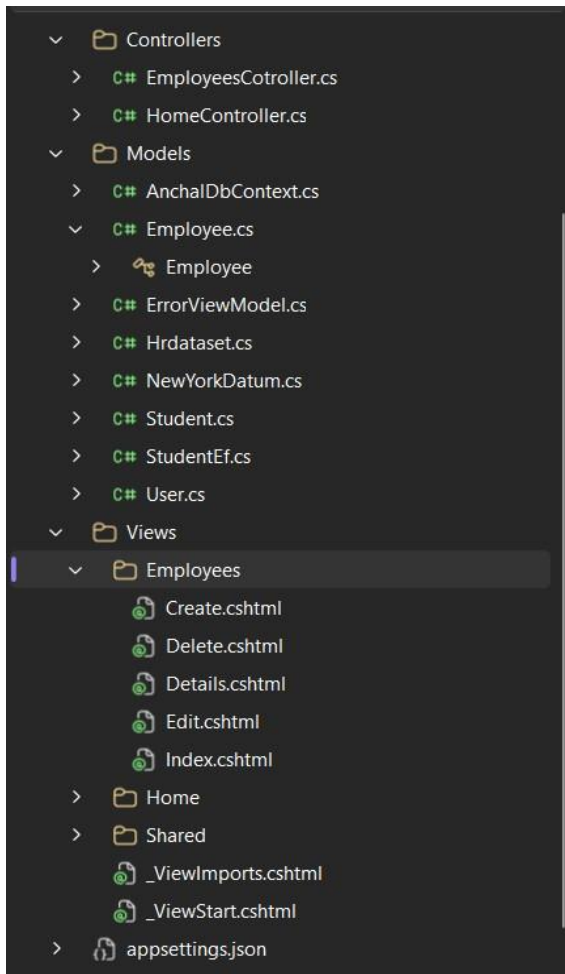
Output

```
PS D:\ANCHAL\AWT Practicals\EmpMvcDbFirst\EmpMvcDbFirst> dotnet aspnet-codegenerator controller -name Employees
Controller -m Employee -dc AnchalDbContext --relativeFolderPath Controllers --useDefaultLayout --referenceScriptLibraries
Building project ...
Finding the generator 'controller'...
Running the generator 'controller'...

Minimal hosting scenario!
Attempting to figure out the EntityFramework metadata for the model and DbContext: 'Employee'
Could not load file or assembly 'Microsoft.EntityFrameworkCore, Version=8.0.11.0, Culture=neutral, PublicKeyToken=adb9793829dda60'. The system cannot find the file specified.
   at Microsoft.VisualStudio.Web.CodeGeneration.ActionInvoker.<BuildCommandLine>b__6_0()
   at Microsoft.Extensions.CommandLineUtils.CommandLineApplication.Execute(String[] args)
   at Microsoft.VisualStudio.Web.CodeGeneration.ActionInvoker.Execute(String[] args)
   at Microsoft.VisualStudio.Web.CodeGeneration.CodeGenCommand.Execute(String[] args)
RunTime 00:00:30.61
PS D:\ANCHAL\AWT Practicals\EmpMvcDbFirst\EmpMvcDbFirst> |
```

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Name: Anchal Singh
Roll No: 119

Employees - EmpMvcDbFirst

localhost:7258

EmpMvcDbFirst

Home

Privacy

Employees

Add New

Id	Name	Email	Salary	Actions
1	Anchal	anchal@gmail.com	1200000.00	EditDetailsDelete

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5. Create partial views

ASP.NET Core Web App (Model-View-Controller) -> EmpMvcApp

Create Model (Emp.cs)

Models folder → Add Class → Emp.cs namespace EmpMvcApp.Models

```
{
    public class Emp
    {
        public int Id { get; set; }
        public string Name { get; set; }
        public string Department { get; set; }
        public decimal Salary { get; set; }
    }
}
```

Controllers → Add → Controller → MVC Controller - Empty EmpController

```
using Microsoft.AspNetCore.Mvc;
using EmpMvcApp.Models;
namespace EmpMvcApp.Controllers
{
    public class EmpController : Controller
    {
        public IActionResult Create()
        {
            return View();
        }
    }
}
```

```
[HttpPost]
public IActionResult Create(Emp emp)
{
    return View(emp);
}

public IActionResult Edit()
{
    var emp = new Emp { Id = 1, Name = "Test", Department = "IT", Salary = 50000 };
    return View(emp);
}
```

```
[HttpPost]
public IActionResult Edit(Emp emp)
{
    return View(emp);
}
```

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}}}

Views → Emp folder

_EmpFormPartial.cshtml

@model EmpMvcApp.Models.Emp

```
<div class="form-group">
<label asp-for="Name"></label>
<input asp-for="Name" class="form-control" />
</div>
<div class="form-group">
<label asp-for="Department"></label>
<input asp-for="Department" class="form-control" />
</div>
<div class="form-group">
<label asp-for="Salary"></label>
<input asp-for="Salary" class="form-control" />
</div>
```

Views/Emp/Create.cshtml

@model EmpMvcApp.Models.Emp

```
<h2>Create Employee</h2>
<form asp-action="Create" method="post">
<partial name="_EmpFormPartial" model="Model" />   <button type="submit" class="btn
btn-primary">Save</button>
</form>
```

Views/Emp/Edit.cshtml

@model EmpMvcApp.Models.Emp

```
<h2>Edit Employee</h2>

<form asp-action="Edit" method="post">
<partial name="_EmpFormPartial" model="Model" />
<input type="hidden" asp-for="Id" />
<button type="submit" class="btn btn-success">Update</button>
</form>
```

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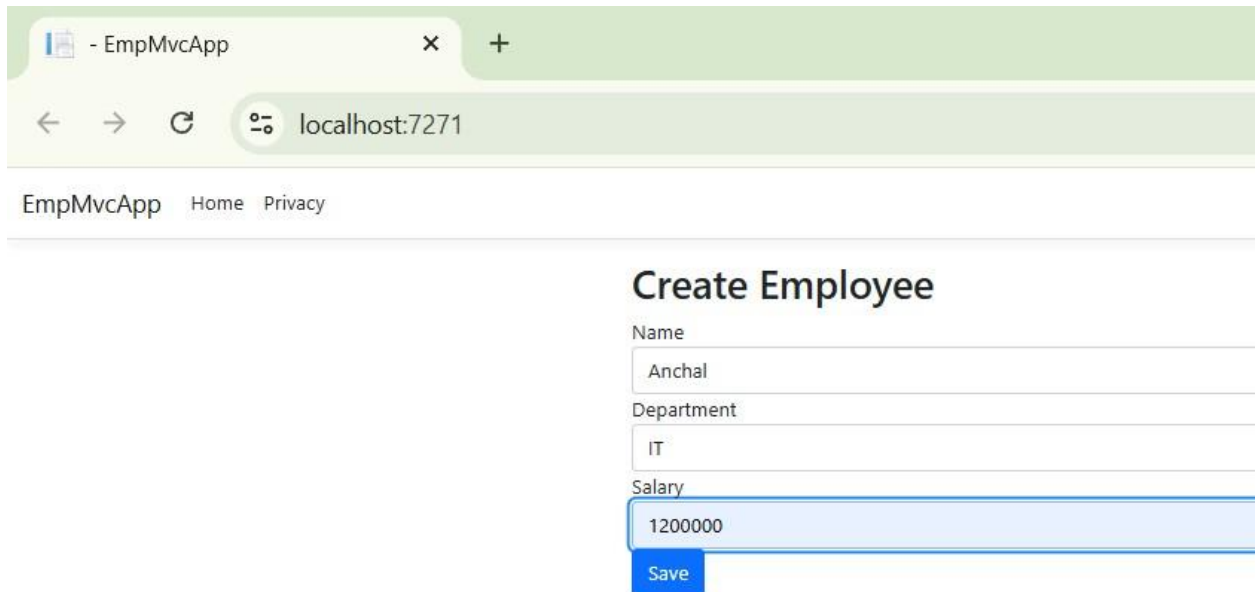
Roll No: 119

Program.cs

```
var builder = WebApplication.CreateBuilder(args); // Add services to the container.
builder.Services.AddControllersWithViews();
var app = builder.Build();
// Configure the HTTP request pipeline.

if (!app.Environment.IsDevelopment())
{
    app.UseExceptionHandler("/Home/Error");
    // The default HSTS value is 30 days. You may want to change this for production scenarios,
    see https://aka.ms/aspnetcore-hsts.
    app.UseHsts(); }
app.UseHttpsRedirection();
app.UseStaticFiles();
app.UseRouting();
app.UseAuthorization();
app.MapControllerRoute(    name: "default",    pattern:
"{controller=Emp}/{action=Create}/{id?}");
app.Run();
```

Output:



The screenshot shows a web browser window with the title '- EmpMvcApp'. The address bar displays 'localhost:7271'. The page content includes a navigation bar with 'EmpMvcApp', 'Home', and 'Privacy' links. The main heading is 'Create Employee'. Below this, there are three input fields: 'Name' with the value 'Anchal', 'Department' with the value 'IT', and 'Salary' with the value '1200000'. A blue 'Save' button is located at the bottom of the form.

Field	Value
Name	Anchal
Department	IT
Salary	1200000